Journal of Water Process Engineering 91 (2026) 110543

Contents lists available at ScienceDirect

Journal of Water Process Engineering

journal homepage: www.elsevier.com/locate/jwpe

# Mechanism-aligned machine learning for membrane fouling prediction and process control in pressure-driven water treatment membranes: A review

Xin Yang, Jundong Zhao, Junzhao Zhuang, Songlin Guan, Chunbo Wang, Kenglin Huang, Lele Liu, Yunan Gao *

*School of Environmental and Chemical Engineering, Foshan University, Foshan, China*

ARTICLE INFO

Editor: Amir Razmjou

*Keywords:*
Membrane fouling
Machine learning
Explainable artificial intelligence
Process control
Water reuse

ABSTRACT

Pressure-driven membrane processes are essential for water reclamation, drinking-water treatment, industrial wastewater reuse and resource recovery. However, their long-term stability is limited by membrane fouling, which is governed by hydraulic history, influent variability, organic-inorganic interactions, biofilm growth and localized scaling. This review develops a mechanism-aligned machine-learning framework for fouling prediction and process control in ultrafiltration, nanofiltration and reverse osmosis systems. The framework links dominant fouling mechanisms with data structures, model architectures and practical control actions. Gated recurrent unit models are discussed for ultrafiltration systems with high-frequency hydraulic signals and rapid cake-layer development, where they can support adaptive flux regulation, air-scouring adjustment and backwash scheduling. Regularized XGBoost models are evaluated for nanofiltration systems with multicollinear water-quality variables and coupled organic-inorganic fouling, where they can guide pretreatment optimization, coagulant dosing and chemical-dose reduction. Hybrid convolutional neural network-gated recurrent unit models combined with SHAP are examined for reverse osmosis systems with spatially heterogeneous biofouling and scaling, where they can support early-warning diagnosis, pH correction, antiscalant control, targeted cleaning and biocide zoning. The review also discusses benchmarking, SCADA noise, data sparsity, fallback control, edge deployment and operator interpretability. Future work should prioritize plant-scale validation, physics-informed learning, digital twins and federated learning to support safer and more deployable artificial intelligence in water-treatment facilities.

## Abbreviations

| Abbreviation | Full term | Abbreviation | Full term |
|---|---|---|---|
| AI | Artificial intelligence | MWCO | Molecular-weight cut-off |
| ANN | Artificial neural network | NF | Nanofiltration |
| APAM | Anionic polyacrylamide | NOM | Natural organic matter |
| ARIMA | Autoregressive integrated moving average | NTU | Nephelometric turbidity unit |
| ATP | Adenosine triphosphate | OCT | Optical coherence tomography |
| CAPEX | Capital expenditure | OPEX | Operating expenditure |
| CART | Classification and regression tree | PCA | Principal component analysis |
| CIP | Clean-in-place | PIML | Physics-informed machine learning |
| CNN | Convolutional neural network | PINN | Physics-informed neural network |
| CNN-GRU-SHAP | CNN-GRU model with SHAP interpretation | PLC | Programmable logic controller |
| DAE | Denoising autoencoder | RF | Random forest |
| DOC | Dissolved organic carbon | RMSE | Root mean square error |
| DOM | Dissolved organic matter | RNN | Recurrent neural network |
| DSPM-DE | Donnan Steric Pore Model with Dielectric Exclusion | RO | Reverse osmosis |
| eDNA | Extracellular DNA | SCADA | Supervisory control and data acquisition |

(continued on next column)

(continued)

(continued on next page)

* Corresponding author.
*E-mail address:* gaoyunan@fosu.edu.cn (Y. Gao).

https://doi.org/10.1016/j.jwpe.2026.110543
Received 20 May 2026; Received in revised form 24 June 2026; Accepted 7 July 2026
Available online 17 July 2026
2214-7144/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(*continued*)

| | | | |
|---|---|---|---|
| EPS | Extracellular polymeric substances | SEC | Specific energy consumption |
| EWMA | Exponentially weighted moving average | SDI | Silt Density Index |
| FP32 | 32-bit floating-point precision | SHAP | SHapley Additive exPlanations |
| GNN | Graph neural network | SVM | Support vector machine |
| GRU | Gated recurrent unit | SVR | Support vector regression |
| INT8 | 8-bit integer precision | TMP | Transmembrane pressure |
| IPC | Industrial PC | UF | Ultrafiltration |
| KNN | K-nearest neighbor | $UV_{254}$ | Ultraviolet absorbance at 254 nm |
| LSTM | Long short-term memory | VAE | Variational autoencoder |
| MCAR | Missing completely at random | XAI | Explainable artificial intelligence |
| MIT | Methylisothiazolinone | XGBoost | Extreme gradient boosting |
| ML | Machine learning | ZLD | Zero-liquid discharge |
| MNAR | Missing not at random | | |

## 1. Introduction

Driven by water scarcity, stricter discharge limits and the need for fit-for-purpose reuse, pressure-driven membrane technologies have become important unit operations in modern water process engineering [1]. These processes form a complementary separation hierarchy for complex water matrices [2]. Ultrafiltration (UF) is widely used in municipal wastewater reclamation and drinking-water pretreatment. Nanofiltration (NF) supports selective ion separation, organic-matter control and resource recovery from industrial streams. Reverse osmosis (RO) provides a dense separation barrier for high-salinity wastewater reuse, brackish-water treatment and polishing steps in advanced treatment trains. Despite these advantages, stable long-term operation remains limited by membrane fouling. Fouling shortens membrane service life, increases cleaning frequency, raises specific energy consumption (SEC) and can reduce the reliability of water-treatment facilities [3].

Importantly, membrane performance decline is not a singular or static event. Instead, it is a highly complex, non-linear and spatiotemporal phenomenon [4]. This process is driven by the synergistic interactions among physical pore blockage, organic adsorption, inorganic scaling and microbial degradation [5]. Across the membrane separation hierarchy, the dominant fouling mechanisms differ substantially in both physical characteristics and temporal evolution. This pronounced heterogeneity makes conventional fouling mitigation strategies fundamentally inadequate. These strategies mainly rely on deterministic mathematical models, such as classical Hermia blocking laws and solution-diffusion derivations [6]. They also depend on static influent water quality indicators, such as the Silt Density Index (SDI). In addition, reactive clean-in-place (CIP) protocols are typically initiated only after severe flux decline has already occurred [7]. Full-scale membrane operations often face dynamic and high-frequency influent fluctuations. Under such conditions, traditional experience-based heuristics cannot effectively decouple complex cross-correlations. Consequently, membrane systems often suffer from suboptimal hydraulic control, excessive chemical consumption and accelerated irreversible degradation of polyamide active layers [8].

To overcome the limits of deterministic modeling, computational tools have been increasingly used in membrane engineering. This trend is moving the field from empirical approximation toward data-assisted process control [9]. Advanced algorithms can extract non-linear patterns from heterogeneous operational datasets. They can also support dynamic hydraulic optimization and early-warning diagnostics [10]. However, a clear analytical gap remains. Many studies apply neural networks or ensemble trees across membrane systems without sufficient mechanistic distinction. They often do not explain why a given model is appropriate for UF, NF, or RO fouling [11]. Broad membrane-technology reviews have summarized sustainable water-treatment applications and general performance limitations, including fouling, but have only limited or indirect discussion of ML-enabled control [12]. Reviews focused on dynamic fouling characterization have clarified the spatial and temporal evolution of deposits in pressure-driven systems, whereas AI model selection and operational decision translation are not their central objectives [13]. AI-oriented reviews have moved the field closer to data-driven fouling prediction, but their comparison is often organized around algorithm categories or prediction accuracy. For example, AI-based fouling-prediction reviews emphasize ANN and related models while treating fouling factors mainly as input variables [14]. Reviews of ML for membrane-separation performance in wastewater treatment summarize common algorithms and performance indicators, but fouling is usually one of several process issues rather than the organizing mechanism [15]. Process-specific reviews on RO and NF provide deeper discussion of scaling, biofouling, separation performance and AI/ML applications, but they remain largely confined to one membrane level and do not systematically connect UF, NF and RO fouling pathways with different model-control requirements [16,17].

A stronger framework is therefore needed. It should align each mathematical architecture with the relevant physicochemical boundary conditions. It should also provide explanations that operators can use before changing hydraulic or chemical setpoints. To address this gap, this review develops a mechanism-driven framework for membrane fouling prediction and process control. The framework links predictive accuracy with physical interpretability and operational usefulness. It first clarifies representative application scenarios of UF, NF and RO systems [18]. It then compares the dominant fouling mechanisms across these separation levels and explains how these mechanisms shape data structures and control objectives. On this basis, the review evaluates three representative model families. Gated recurrent unit (GRU) models are discussed for rapid UF cake-layer development [19]. XGBoost is discussed for NF systems with coupled organic-inorganic water-quality variables [20]. Hybrid CNN-GRU models are discussed for RO systems with spatially heterogeneous biofouling and scaling [21]. This article is a narrative and critical review rather than a formal meta-analysis. It does not pool effect sizes across studies. Instead, it organizes the literature by membrane process, dominant fouling mechanism, data type, validation approach and control objective [22]. Four questions guide the analysis: which fouling mechanism dominates the process? which data structure is available? which model architecture fits that structure? and which operational decision the model can support? This logic helps avoid the generic use of black-box algorithms and makes the engineering value of each model easier to evaluate [23]. Finally, the review examines explainable artificial intelligence (XAI), fallback control and edge deployment as requirements for safer use of AI in digital-twin-enabled water infrastructure.

## 2. Literature search and selection methodology

This review was designed as a critical narrative review with a transparent and reproducible literature-selection procedure rather than as a quantitative meta-analysis. The literature search covered peer-reviewed articles, reviews and selected methodological papers published from January 2010; earlier landmark studies were retained only when they were needed to define classical fouling mechanisms, membrane-process terminology or model-theory foundations. The search was conducted in Scopus, Web of Science Core Collection, ScienceDirect, Google Scholar, IEEE Xplore and PubMed. Search strings combined three term groups: membrane processes (“ultrafiltration”,

"nanofiltration", "reverse osmosis" and "pressure-driven membrane"), fouling mechanisms ("membrane fouling", "cake layer", "pore blocking", "organic fouling", "biofouling" and "scaling") and data-driven control methods ("machine learning", "deep learning", "XGBoost", "GRU", "CNN", "SHAP", "explainable artificial intelligence", "process control" and "digital twin"). Backward and forward citation tracking was then performed for highly cited membrane-fouling reviews, recent AI-assisted membrane studies and key papers on interpretable machine learning.

The initial database search identified 1822 records: Scopus ($n$ = 548), Web of Science Core Collection ($n$ = 462), ScienceDirect ($n$ = 344), Google Scholar (first 250 relevance-ranked records), IEEE Xplore ($n$ = 118) and PubMed ($n$ = 100). Citation tracking added 41 records, giving 1863 records before deduplication. After 502 duplicate records were removed, 1361 records remained for title and abstract screening. At this stage, 877 records were excluded because they were outside pressure-driven membrane systems, did not address fouling, lacked a data-driven or control-relevant component, or focused only on material synthesis without operational interpretation. The remaining 484 full-text articles were assessed in detail. Of these, 210 were excluded because they lacked sufficient process conditions, did not report fouling- or control-related outputs, provided no model-validation information, were not relevant to UF/NF/RO water-treatment operation, or substantially overlapped with a more complete later publication. Finally, 274 publications were retained and cited in the review.

Studies were included when they met at least one of the following criteria: (i) they examined fouling formation, fouling diagnosis, flux decline, permeability loss, transmembrane-pressure evolution, salt-rejection deterioration or cleaning response in UF, NF or RO systems; (ii) they used machine-learning, deep-learning, hybrid mechanistic-data-driven, explainable-AI or control-oriented modeling approaches relevant to membrane operation; (iii) they reported operating variables, sensor signals, water-quality descriptors, imaging or spectroscopic data that could be linked to fouling signatures; or (iv) they provided methodological foundations for model validation, uncertainty handling, feature attribution, deployment latency, edge implementation or safe process control.

Studies were excluded when they focused exclusively on non-pressure-driven separation, lacked a water-treatment or membrane-operation context, reported only qualitative claims without usable technical detail, duplicated information already covered by a more complete source, or could not be connected to the mechanism-to-control framework developed in this review. Because the aim was to build a mechanism-aligned synthesis, the final analysis prioritized evidence that explicitly linked fouling mechanisms, measurable data signatures, model outputs and actionable operational decisions.

## 3. Fouling mechanisms and data signatures in pressure-driven membrane processes

To deploy predictive algorithms in water-treatment facilities, the model architecture should reflect both the physical fouling process and the available operational data. This section therefore examines the engineering context of pressure-driven membrane technologies. It identifies the main physicochemical failure points, the associated data signatures and the control actions that are realistic for each separation level.

### 3.1. Application scope and process-control value

Pressure-driven membrane technologies are used across a wide range of water-treatment, wastewater-reuse and resource-recovery scenarios. Their engineering value is shaped by influent composition, regulatory targets, hydraulic constraints and available plant infrastructure. UF is widely used as a robust barrier for suspended solids, colloids and pathogens in drinking-water production and wastewater reclamation. NF is often applied where selective ion separation, organic-matter control or resource recovery is required. RO contains a compact polyamide active layer and provides the densest pressure-driven separation barrier for high-salinity wastewater reuse, brackish-water treatment and advanced polishing. Table 1 summarizes quantitative separation metrics, operating ranges, warning thresholds, sampling intervals and deployment-related model metrics; qualitative input variables and mechanism-control interpretations are discussed below.

As shown in Table 1, UF, NF and RO require different model-control pairings because their measurable data streams and operational risks are not identical. UF models are usually driven by high-frequency transmembrane pressure (TMP), flux, turbidity, particle-count, temperature and backwash-history records. These signals reflect rapid particle deposition, colloid accumulation and cake-layer compaction under shock loads, so sequence models or anomaly-detection modules can support flux correction, air-scouring adjustment and backwash scheduling [43]. NF models are usually built from pressure, flux, conductivity, DOC, $UV_{254}$, pH and dosing records. These variables describe steric exclusion, electrostatic effects and dissolved organic matter (DOM)-metal interactions, so regularized tree ensembles can support coagulant, pH and antiscalant-dose optimization [44]. RO models combine pressure, flux, conductivity, pH and antiscalant records with optical coherence tomography (OCT) or hyperspectral fouling images when available; because localized biofouling, scaling and chemical-cleaning stress are strongly spatiotemporal, multimodal or hybrid models are more suitable for early-warning, targeted chemical dosing and CIP-timing decisions [45]. Thus, the key issue for AI-assisted process control is not to rank these processes as good or poor, but to match each process with the data streams, fouling mechanisms, validation methods and intervention strategies that are most relevant to its engineering role.

### 3.2. Membrane fouling mechanisms and data signatures

As shown in Fig. 1, the cross-sectional fouling schematic provides a standardized visual basis for comparing fouling components across UF, NF and RO systems. The figure uses three aligned panels corresponding to UF, NF and RO. In each panel, the feed side, membrane layer, fouling components and permeate side are labelled consistently. For UF, the labelled components include colloidal particles, suspended solids, EPS/macromolecular organics, pore blocking, reversible cake layer and

**Table 1**
Quantitative operating ranges, warning thresholds and deployment metrics for representative pressure-driven membrane processes.

| Data item | UF | NF | RO | Ref. |
|---|---|---|---|---|
| Nominal separation metric | 0.01–0.10 μm | 200–1000 Da | >95–99% | [24,25] |
| Operating pressure or TMP | 20–100 kPa | 5–20 bar | 10–25 bar; 40–70 bar | [26,27] |
| Flux or solute-rejection range | 30–150 LMH | 50–95% | >95–99% | [28–30] |
| Online sampling interval | 1–60 s | 1–15 min | 1–5 min | [31,32] |
| Primary fouling warning threshold | +5–20 kPa; >50 NTU; −10-20% | −5-20% | −5-15%; +10–15 kPa | [33,34] |
| Control KPI trigger | <80–100 kPa | −10-30% | −10-15%; +10–15 kPa | [35,36] |
| Offline training time | 0.1–2 h | 0.01–1 h | 1–48 h | [37–39] |
| Online inference latency | 2–20 ms | <1–10 ms | 20–200 ms | [41] |

Note: The ranges in Table 1 are approximate engineering values rather than fixed benchmarks. Offline training is assumed to be performed on standard workstations or edge servers, whereas online inference is assumed to run on local IPCs, PLC-connected gateways or compressed edge devices. The nominal separation metrics refer to UF pore size, NF molecular-weight cut-off and RO salt rejection, respectively. Actual values may vary with feed composition, membrane configuration, dataset size, sampling frequency, model-tuning strategy and available computing resources.

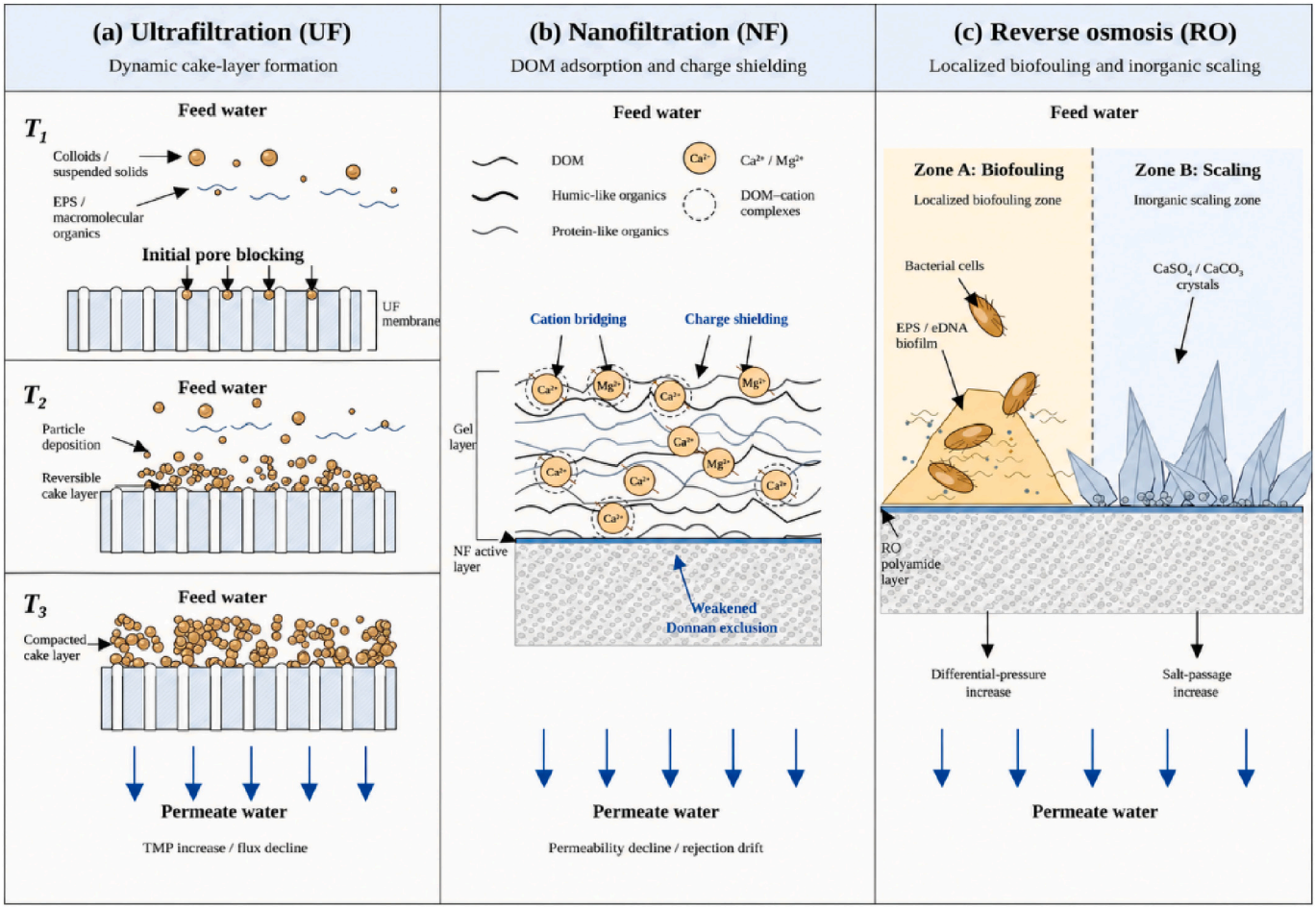



**Fig. 1.** Standardized cross-sectional fouling components and mechanisms in UF (a), NF (b) and RO (c) systems.

compacted cake layer [46]. For NF, the labelled components include DOM, humic-like organics, protein-like organics, multivalent cations, DOM–cation complexes, gel layer, charge shielding and weakened Donnan exclusion [47]. For RO, the labelled components include bacterial cells, extracellular polymeric substances (EPS) /eDNA biofilm, inorganic scale crystals, $CaSO_4/CaCO_3$ scaling zones and localized heterogeneous fouling. These standardized labels make the figure consistent with the following mechanistic discussion and help readers compare the dominant fouling pathways among UF, NF and RO systems.

### 3.2.1. *UF: Dynamic cake-layer formation*

Fig. 1a illustrates the sequential evolution of UF fouling from initial pore blocking to cake-layer compaction. At the initial stage, physical pore interception is the dominant fouling mechanism. This stage is soon followed by the formation of a macroscopic cake layer [49]. The whole process is highly dynamic and is accompanied by high-frequency hydraulic fluctuations. For example, algae-laden surface water or stormwater-impacted municipal wastewater can cause rapid particle deposition [50]. These deposited materials include colloidal particles, EPS and macromolecular organics. As a result, fouling develops quickly and triggers a rapid increase in TMP. This increase can occur within short operational windows [51]. The deterioration may progress from reversible pore shielding to irreversible cake-layer compaction. Eventually, a composite fouling architecture is formed. Fundamentally, this process is controlled by hydrodynamic shear forces and localized concentration polarization [52].

### 3.2.2. *NF: DOM adsorption and donnan shielding*

In NF systems, as shown in Fig. 1b, the fouling trajectory is shaped by several interacting mechanisms. These include steric hindrance, hydrophobic affinity, and Donnan electrostatic interactions. DOM plays a key role in initiating interfacial resistance. More specifically, low-molecular-weight protein-like substances are major contributors [53]. Medium-to-high-molecular-weight humic compounds also act as important fouling instigators. These specific DOM fractions are especially influential. They account for up to 90.17% of the variance in NF fouling trajectories [54].

The coexistence of multivalent metal cations, such as $Ca^{2+}$ and $Mg^{2+}$, with humic substances creates a serious engineering challenge [55]. The Donnan Steric Pore Model with Dielectric Exclusion (DSPM-DE) can help explain this interaction at the molecular scale [56]. In this framework, multivalent cations cross-link with specific natural organic matter (NOM) fractions. This process changes the effective volumetric charge density ($X_d$) in the nanopores. It may also alter the local dielectric constant ($\epsilon_p$) [57]. These organic-inorganic complexes can mask the negative functional groups on the polyamide surface. Carboxylate groups are especially important. In some cases, charge inversion may occur.

This molecular-level charge shielding severely weakens the Donnan potential at the membrane-solution interface. As a result, electrostatic repulsion fails prematurely [58]. At the macroscopic scale, this microscopic interaction causes a highly non-linear and asymptotic decline in permeability. At the same time, multivalent ion selectivity undergoes a systemic collapse. The gelation process is driven by strong synergistic interactions. These interactions generate highly multidimensional operational datasets with extreme multicollinearity. This problem is reflected in the strongly overlapping variance patterns among fluctuating pH, divalent cation concentrations, and $UV_{254}$ absorbance [59].

This complex and strongly coupled data matrix creates major challenges for traditional deterministic models. Therefore, regularized high-dimensional gradient boosting frameworks, such as XGBoost, are needed. These frameworks can decouple the feature spaces and identify the primary fouling vectors [60].

#### 3.2.3. RO: Scaling and biofouling

RO membranes contain compact polyamide active layers and therefore provide the highest-density pressure-driven separation barrier [61]. This structural advantage supports high salt rejection, but it also makes long-term operation sensitive to inorganic scaling and biological degradation [62]. Fig. 1c highlights the spatiotemporal heterogeneity of RO fouling. In these systems, different failure modes may appear in localized micro-environments.

During brackish-water desalination or high-salinity industrial wastewater treatment, supersaturated ions often crystallize on the membrane surface [63]. These crystals then form inorganic scales. Dense gypsum scales formed by calcium and sulfate are a typical example. This pathway corresponds to the scaling zone in Fig. 1. Such crystallization can account for more than 90% of the total inorganic fouling mass [64]. At the same time, localized biofouling creates an active and continuously growing threat. Initial biofilm colonization strongly depends on the physicochemical properties of the conditioning layer. Its polarity and acid-base free energy are particularly important [65]. After initial bacterial attachment, microorganisms secrete a complex matrix of EPS. This matrix mainly consists of highly hydrated polysaccharides, structural proteins, and extracellular DNA (eDNA) [66].

This EPS assembly forms a strong three-dimensional gel scaffold. It acts as both a physical and biochemical shield. It slows the diffusion of standard CIP agents, such as sodium hypochlorite and non-oxidizing biocides. It can also reduce their toxicity. As a result, the bacterial community becomes more resistant to environmental stressors [67]. Consequently, biofilm formation is highly insidious and non-linear. It is also marked by strong spatiotemporal heterogeneity [68]. When conventional supervisory control and data acquisition (SCADA) sensors detect increased TMP or reduced permeate flow, the biofilm is usually already mature. At this stage, the biofilm is encased in EPS and becomes highly recalcitrant. Therefore, reactive CIP protocols are often ineffective.

To overcome this limitation, biofilm evolution must be monitored at the nascent stage. Localized EPS matrix densification can produce distinct optical refractive-index variations [69]. Dynamic changes in the hydration layer can also generate morphological thickness microtextures. These features can be detected by non-invasive optical monitoring techniques, including OCT and hyperspectral imaging [70]. These complex biological transformations can be mapped into high-resolution spatial feature tensors. Based on these tensors, convolutional neural networks (CNN) can quantitatively extract subtle optical perturbations through multidimensional convolutional kernels [71]. By directly linking physical microbiology with computer-vision features, advanced predictive models can function as digital microscopes for early biofouling diagnosis [72]. This approach provides a rigorous scientific basis for diagnosing and intercepting early biofouling. It also allows intervention before irreversible macroscopic hydraulic deterioration occurs [73].

For non-invasive RO fouling monitoring, OCT and hyperspectral imaging should be treated as complementary diagnostic tools rather than as equivalent sensors [74]. OCT is mainly suited to depth-resolved structural diagnosis because it can reconstruct cross-sectional or three-dimensional information on fouling-layer thickness, biofilm architecture and cake or scale morphology. However, its field of view is limited, optical access to the membrane surface is required and signal quality may decline in highly turbid feed channels or modules without optical windows [75]. Hyperspectral imaging is more suitable for larger-area chemically informative screening because wavelength-dependent spectral signatures can reflect organic matter, biofilm pigments, hydration-state changes and some mineral deposits. However, it produces high-dimensional spectral cubes that require stable illumination, calibration, feature compression and data-storage capacity [76]. Therefore, OCT is better positioned for pilot-scale studies, sentinel modules or selected high-risk RO trains, whereas hyperspectral imaging is more appropriate for offline or near-line diagnosis and targeted modules coupled with edge preprocessing before CNN-GRU modeling [77].

### 3.3. Limits of deterministic models and need for data-driven control

Conventional fouling mitigation strategies rely heavily on static influent water-quality guidelines, such as SDI thresholds. They also depend on deterministic mass-transfer equations. In addition, passive chemical cleaning is often used. However, this cleaning is usually triggered only after severe flux decline has already occurred [78]. These empirical heuristics are no longer sufficient because they cannot fully address the dynamic realities of modern water-treatment plants.

For example, traditional engineering guidelines often assume that high volumetric recovery rates directly lead to high fouling risks. However, this assumption is too simplistic. It may unnecessarily limit system water recovery efficiency [79]. Conventional chemical interventions can also create secondary problems. Low concentrations of non-oxidizing biocides, such as methylisothiazolinone (MIT), may activate quorum sensing pathways in disinfectant-resistant bacteria [80]. This activation can promote aggressive biofilm formation. It may also increase biomass by up to 180% [81]. At a fundamental level, deterministic mathematical models struggle to resolve deep non-linear cross-correlations. These correlations involve fluctuating pH, temperature, complex DOM fractions and hydrodynamic shear forces. Their complexity becomes even more pronounced during prolonged operational cycles [82].

AI technologies offer strong potential for addressing this industry-wide bottleneck. However, the blind application of generic algorithms is unlikely to succeed. The physical mechanistic differences discussed above must be properly considered [83]. Different fouling mechanisms require different AI capabilities. Macroscopic cake-layer accumulation requires rapid temporal response [84]. Charge masking in gel layers involves extreme multicollinearity. It therefore requires high-dimensional feature decoupling [85]. Localized biofilm outbreaks require multimodal spatial processing. This processing is needed to capture their extreme spatiotemporal evolution [86]. The following sections establish a unified framework. This framework aims to align frontier predictive architectures with distinct physical mechanisms in a precise mathematical manner.

## 4. Predictive architectures and mechanistic alignment

Across the pressure-driven membrane hierarchy, fouling prediction requires models that are consistent with the underlying process dynamics. A useful model should not only improve statistical accuracy, but also support a feasible and bounded control action, such as adaptive backwashing, pretreatment-dose adjustment, pH correction, antiscalant control or targeted cleaning [87]. To make this logic clearer, Fig. 2 summarizes the mechanism-to-control pathway rather than only presenting a detailed algorithmic illustration. The figure follows five sequential layers: dominant fouling mechanism, measurable data signature, model selection, operator-facing model output and practical control action.

As illustrated in Fig. 2, the mechanism-aligned framework should be interpreted as a control-oriented decision pathway rather than a simple comparison of model architectures. The pathway begins with the dominant fouling mechanism, converts this mechanism into measurable data signatures, selects a model according to the available data structure, and finally translates the model output into bounded operational

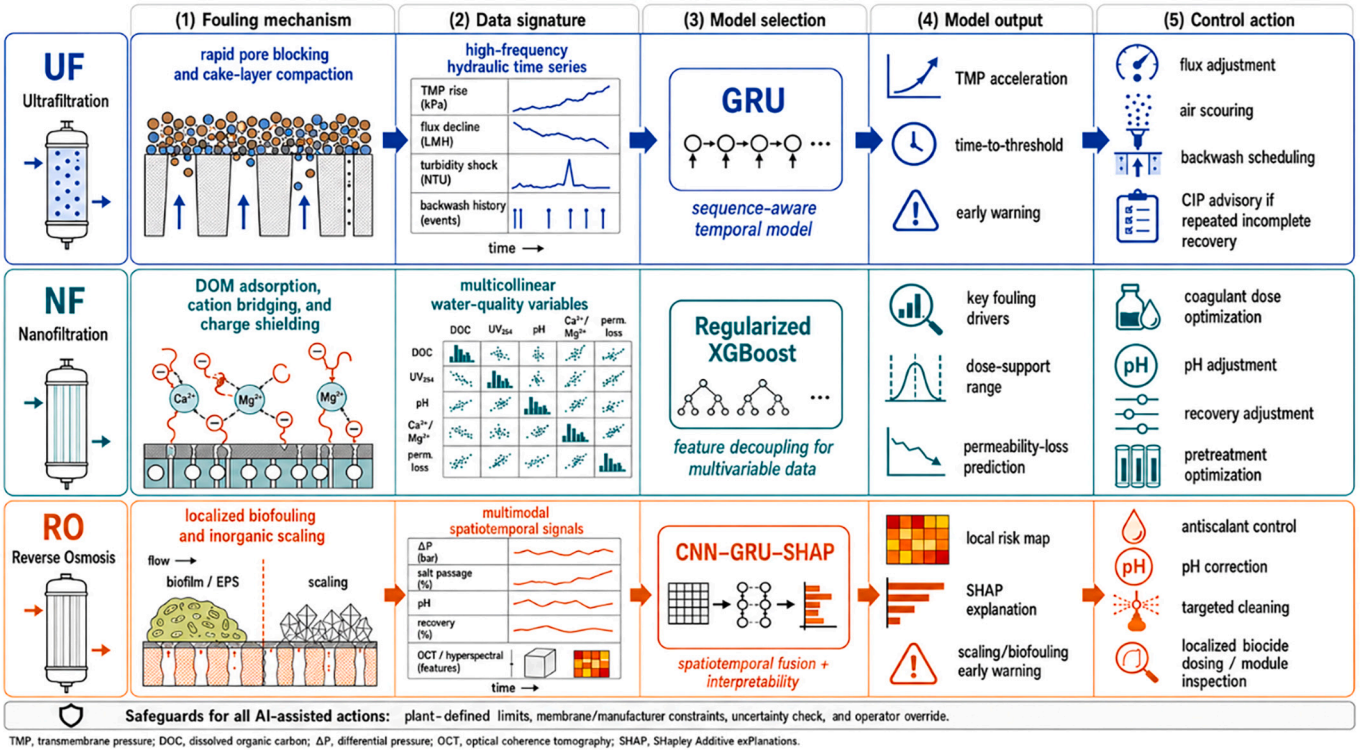



**Fig. 2.** Proposed mechanism-to-control pathway for AI-assisted membrane-fouling management.

actions. This structure makes the role of each algorithm clearer. GRU models are mainly suitable for high-frequency UF hydraulic sequences because they can predict TMP acceleration and support flux, air-scouring and backwash adjustment. Regularized XGBoost is suitable for NF systems with multicollinear water-quality variables because it can identify dominant organic-inorganic fouling drivers and support pretreatment or dosing decisions. CNN-GRU-SHAP models are most useful for RO systems when spatially resolved images or multimodal data are available, because they can connect localized scaling or biofouling signatures with early-warning diagnosis, SHAP-based explanation and targeted cleaning or chemical control.

The figure links dominant fouling mechanisms, measurable data signatures, model selection, model outputs and operator-facing control actions for UF, NF and RO systems. For UF, rapid pore blocking and cake-layer compaction generate TMP slope, flux decline, turbidity and backwash-history signals, which can be processed by GRU models to predict TMP acceleration and guide flux regulation, air scouring and backwash scheduling [88]. For NF, DOM adsorption, cation bridging and charge shielding generate multicollinear water-quality and permeability signals, which can be analyzed by regularized XGBoost to identify key fouling drivers and support coagulant dosing, pH optimization and recovery adjustment [89]. For RO, localized scaling and biofouling generate differential-pressure, salt-passage, pH, recovery and imaging-based signatures, which can be integrated by CNN-GRU-SHAP models to provide spatial risk diagnosis, SHAP-based explanations, antiscalant or pH correction, targeted cleaning, biocide zoning and module inspection [90]. All actions should be constrained by plant-defined limits, membrane-manufacturer recommendations and operator override.

### *4.1. Practical interpretation and implementation of the mechanism-aligned framework*

The UF–GRU, NF–XGBoost and RO–CNN–GRU–SHAP links proposed in this review should be interpreted as representative guidelines rather than fixed model-allocation rules. In practical membrane plants, model selection depends on data availability, sampling frequency, sensor modality, dataset size, prediction horizon and the intended control objective [91]. For example, XGBoost can also be highly useful for RO plant datasets when the available inputs are structured SCADA variables, such as pressure, flow rate, recovery, conductivity, pH, temperature, antiscalant dose and normalized salt passage. Conversely, recurrent models can be relevant to NF systems when the target is time-dependent permeability loss, salt-rejection drift or cleaning demand under high-frequency operation. Similarly, CNN-based components are justified only when image, OCT, hyperspectral or spatially resolved sensor data are available [93]. Therefore, the proposed framework should be used as a decision pathway. The fouling signature and data structure should be identified first, and the simplest model that satisfies the prediction and control requirement should then be selected.

To make the discussion of fouling mechanisms more operational, each mechanism should be linked to measurable indicators. In UF systems, cake-layer formation and compaction can be tracked using TMP increase rate, normalized flux decline, permeability loss, turbidity, particle counts, backwash interval, air-scouring demand and CIP frequency [94]. In NF systems, organic-inorganic fouling should be connected with normalized permeability, rejection drift, DOC, $UV_{254}$, fluorescence indices, pH, ionic strength and $Ca^{2+}/Mg^{2+}$ concentrations. These variables reflect DOM adsorption, cation bridging and charge shielding [95]. In RO systems, scaling and biofouling should be evaluated using differential pressure, normalized permeate flow, normalized salt passage, conductivity, pH, temperature, recovery, antiscalant dose, calcium or sulfate saturation tendency, ATP/EPS indicators when available, cleaning frequency and specific energy consumption [96]. The prediction target should therefore be defined as a plant-relevant endpoint. Examples include TMP increase over a future filtration cycle, percentage permeability loss, probability of crossing a cleaning threshold, expected time to CIP, or recommended dosing, flux or backwash adjustment [97].

Before any model is linked to control, an implementation workflow is required. Raw plant data should first be synchronized by timestamp and separated into filtration, backwash, chemical-cleaning, shutdown and abnormal-operation periods [98]. Physically impossible readings should

be removed or flagged according to engineering limits [99]. Noisy hydraulic signals may be treated using moving-median, low-pass or Kalman-type filters [100]. However, the filtering window should remain shorter than the expected early-warning horizon so that rapid TMP increases or flux-loss signals are not erased. Missing data should be handled according to gap length and physical plausibility. Short gaps may be filled by interpolation, K-nearest neighbor (KNN) or model-based reconstruction [101]. Long gaps should trigger missingness flags, sample exclusion or fallback to a more conservative model [102]. When possible, variables should be normalized using temperature-corrected permeability, normalized flux and normalized pressure, because raw values may mix true fouling effects with changes in operational setpoints.

Feature construction and validation should also be process-specific. For tabular XGBoost models, candidate inputs should include water-quality variables, hydraulic variables, chemical-dosing records and lagged or rolling descriptors of recent operation [103]. Feature selection can combine engineering judgement, permutation importance and SHAP analysis, while recognizing that feature importance does not by itself prove causality [104]. For GRU or CNN-GRU models, sequence length, prediction horizon and sampling frequency should be selected according to filtration-cycle duration and the required control lead time. Random train splitting should be avoided for temporally correlated plant data. A more defensible strategy is chronological splitting, rolling-origin validation, or independent validation across seasons, influent regimes or membrane modules. Hyperparameters should be tuned only on the validation set, including GRU hidden units, dropout and sequence window [105]; XGBoost tree depth, learning rate, number of trees and L1/L2 penalties [106]; and CNN kernel size, filter number and compression ratio [107]. Model uncertainty should be reported using ensemble predictions, quantile regression or confidence intervals [108]. Finally, model drift should be monitored through residual trends, input-distribution shifts and changes in fouling-response relationships. When drift exceeds a plant-defined tolerance, the model should be recalibrated or retrained before it is allowed to support automatic control [109]. To clarify how these requirements can be implemented in practice, Fig. 3 summarizes a plant-level deployment workflow from data acquisition to safeguarded control and feedback-based retraining.

Fig. 3 further translates the mechanism-aligned modeling concept into a plant-level deployment workflow for membrane-fouling prediction and process control. The workflow starts from heterogeneous plant data streams, including SCADA records, water-quality measurements, OCT or hyperspectral images and cleaning-history information. These raw inputs are first processed through data curation steps, including timestamp synchronization, anomaly flagging, filtering, missing-data handling and normalization. The curated data are then transformed into mechanism-relevant features, such as hydraulic trends, water-quality descriptors, imaging signatures, rolling statistics and cleaning-recovery indicators. This step is important because model predictions are operationally meaningful only when true fouling evolution is separated from sensor noise, operational switching and cleaning-related disturbances.

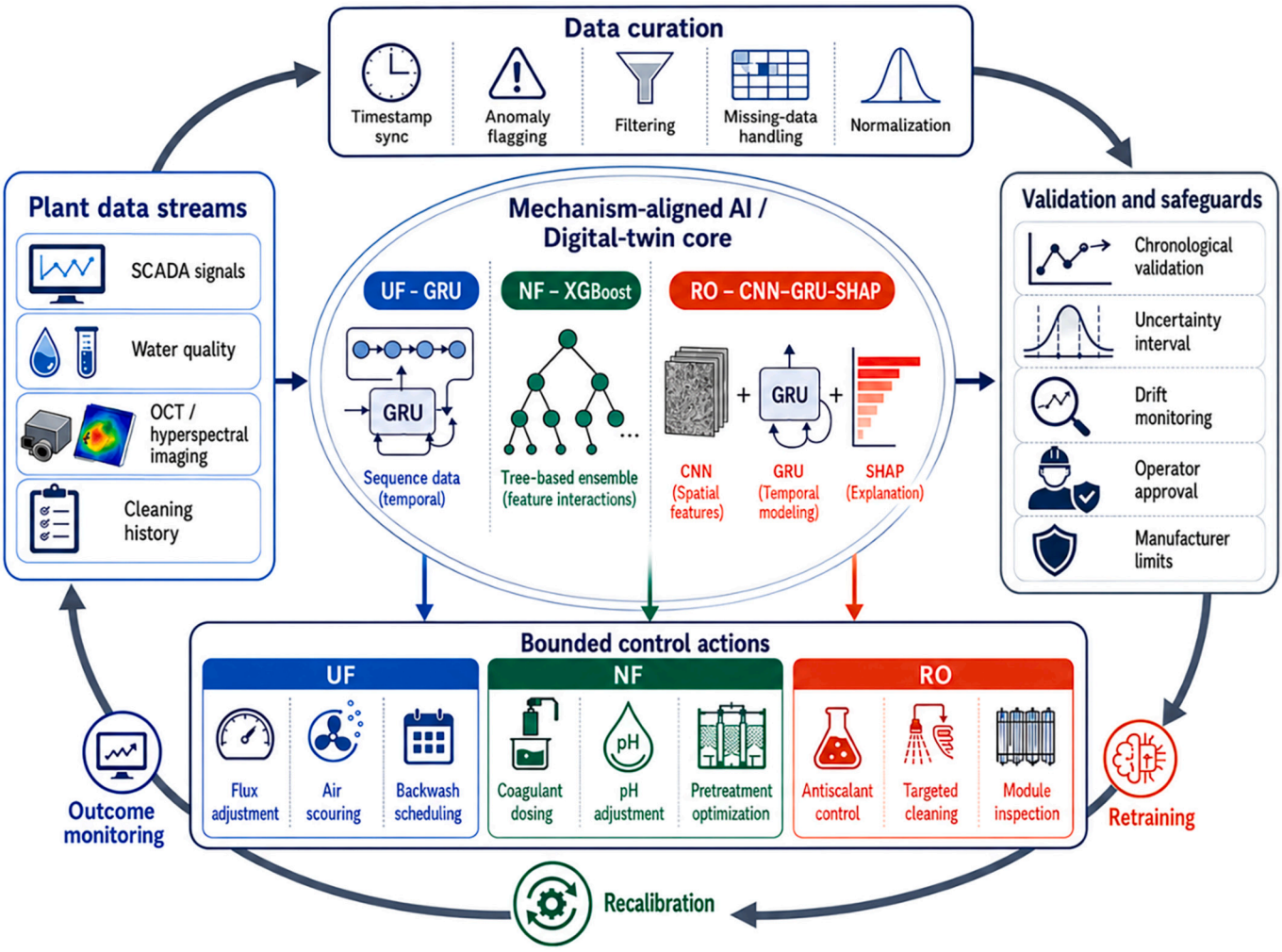



**Fig. 3.** Operational deployment workflow for mechanism-aligned machine learning in membrane-fouling prediction and process control.

The central part of the workflow shows that model selection should follow the dominant fouling mechanism and data structure rather than algorithmic complexity alone. For UF systems, high-frequency hydraulic sequences can be processed by GRU models to support flux adjustment, air-scouring optimization and backwash scheduling. For NF systems, multicollinear water-quality and permeability variables can be handled by regularized XGBoost to guide coagulant dosing, pH adjustment and pretreatment optimization. For RO systems, multimodal spatiotemporal signals can be integrated by CNN–GRU–SHAP models to support antiscalant control, targeted cleaning and module inspection. Before any recommendation is used for process control, the workflow requires chronological validation, uncertainty checking, drift monitoring, operator approval and manufacturer-limit constraints. Therefore, Fig. 3 emphasizes that AI-assisted membrane operation should be implemented as a safeguarded decision-support loop rather than as unconstrained automatic control.

Based on this deployment workflow, the mechanism-aligned framework becomes an operational guide rather than a rigid taxonomy [110]. Its purpose is not to claim that one membrane process must use only one algorithm. Instead, it clarifies which combinations of fouling mechanism, sensor data, model structure and control endpoint are technically defensible [111]. This interpretation is more consistent with full-scale membrane operation, where the best model is determined by instrumentation, data quality, control risk and operator trust.

### 4.2. Translating model outputs into practical process-control actions

For AI-assisted membrane operation, the predicted value itself is not the final engineering product. A model output must be translated into a bounded, auditable and reversible control action [112]. This requirement is particularly important because membrane plants are safety-critical systems. A prediction should therefore be interpreted through three sequential steps: diagnosis, decision support and constrained actuation [113]. First, the model identifies the probable fouling state or the probability of crossing a plant-defined threshold [114]. Second, the prediction is converted into a recommended action, such as flux adjustment, backwash rescheduling, coagulant-dose correction, pH adjustment, antiscalant-dose optimization, cleaning initiation or membrane-condition inspection [115]. Third, the recommended action is checked against operator-defined limits, chemical-safety rules and membrane-manufacturer constraints before implementation [116].

In UF systems, a GRU-based warning of rapid TMP acceleration or flux decline can be converted into hydraulic interventions rather than immediate chemical cleaning [117]. When the predicted TMP slope exceeds a plant-defined warning limit within the next filtration cycle, the controller may shorten the filtration interval, increase backwash frequency, adjust air-scouring intensity or temporarily reduce permeate flux [118]. If repeated warnings are accompanied by incomplete hydraulic recovery, chemically enhanced backwash or CIP can be scheduled [119]. These actions should remain constrained by plant-defined TMP and flux limits, operator override and confirmation from normalized permeability recovery and cleaning history, so that the model output is connected to preventive hydraulic control before irreversible cake-layer compaction occurs.

In NF systems, XGBoost outputs are most useful when they are linked to pretreatment and dosing decisions. If the model attributes permeability loss to elevated $UV_{254}$, DOC, $Ca^{2+}/Mg^{2+}$ concentration or unfavorable pH, the control response may involve adjusting coagulant dose, optimizing pH, reducing instantaneous recovery, or limiting polymer addition to avoid secondary membrane fouling [120]. If the output indicates rejection drift or molecular weight cut-off (MWCO) attenuation risk, the recommended response should be to modify pretreatment and intensify ion-rejection monitoring rather than to apply a fixed chemical correction. The model should not directly prescribe an unrestricted chemical dose; instead, it should recommend a dose range within plant-approved upper and lower limits and verify the recommendation with conductivity, $DOC/UV_{254}$ and ion data. This design allows AI to support chemical optimization while avoiding excessive sludge production, residual coagulant carryover and membrane-surface damage [121].

In RO systems, model outputs should be translated into scaling and biofouling management actions [122]. A predicted increase in localized scaling risk may support pH correction, antiscalant-dose adjustment within product and manufacturer limits, recovery reduction or brine-channel flushing, but the action should be checked against saturation indices and online pH data [123]. A predicted biofouling risk may support targeted cleaning, localized biocide dosing, flow redistribution or inspection of modules with abnormal pressure-drop development, preferably confirmed by pressure-drop trends, ATP/EPS indicators or OCT/hyperspectral imaging when available [124]. Membrane autopsy or replacement should be considered only when normalized permeability, salt rejection or differential pressure cannot be restored after repeated cleaning and when multi-cycle evidence indicates persistent irreversible deterioration. Therefore, the practical role of AI is not to replace operator judgement, but to provide earlier, more specific and more defensible control recommendations [126].

Across UF, NF and RO systems, the operational value of ML outputs depends on whether they can be converted into specific, bounded and verifiable control actions [127]. A practical control layer should therefore distinguish the model signal, the engineering interpretation, the possible intervention and the required safeguard before actuation [128]. This logic avoids the direct use of an unconstrained prediction for automatic control and emphasizes that every AI-assisted action should remain subject to plant limits, manufacturer recommendations, multi-sensor confirmation and operator approval.

### 4.3. UF: Sequence-aware GRU for hydraulic control

#### 4.3.1. Architectural advantages of GRU

For UF systems, performance degradation is mainly driven by physical pore blockage and dynamic cake-layer compaction. These processes are rapid and cumulative [129]. Therefore, standard single-layer ML algorithms, such as Support Vector Machines (SVM) or Random Forests (RF) are structurally inadequate [130]. These static models usually assume that observations at discrete time steps are statistically independent [131]. As a result, they ignore the historical and cumulative stress vectors that drive fouling evolution.

The GRU is a streamlined variant of Recurrent Neural Network (RNN). It is also computationally efficient. This architecture can effectively overcome the above limitation [132]. The structural advantage of the GRU framework comes from its internal mathematical gating mechanisms. These mechanisms include the Update Gate and the Reset Gate [133]. The update gate regulates information flow in a dynamic manner. It evaluates how much historical memory should be retained. This allows the network to capture continuous temporal dependencies over extended periods [134]. In contrast, the reset gate enables the model to discard redundant historical noise.

From a computational mathematics perspective, this streamlined gating mechanism effectively mitigates the vanishing gradient problem [135]. This problem is common in traditional RNN when they process long sequences [136]. More importantly, the GRU shows a clear computational advantage over the traditional Long Short-Term Memory (LSTM) network [137]. This advantage becomes especially important in full-scale municipal water facilities with high-frequency SCADA data sampling. This superiority is rooted in the GRU architecture. Unlike LSTM, GRU deliberately omits the output gate. As a result, it significantly reduces the total number of trainable tensor parameters [138]. In high-frequency hydraulic control, this simplified architecture offers several benefits. It allows the GRU to maintain high temporal predictive accuracy [139]. It also accelerates backpropagation training. In addition, it reduces the memory footprint during inference [140]. This mathematical simplicity is the key reason why GRU is suitable for deployment on constrained local Industrial PCs (IPCs). It can also meet

the low latency requirements of real-time, variable-frequency backwash control [141].

#### 4.3.2. Alignment with UF fouling and control actions

As shown in Fig. 2, the intrinsic design of GRU aligns well with the high-frequency and dynamic characteristics of UF fouling. Municipal UF plants may experience sudden perturbations. These include seasonal algal blooms and massive turbidity shocks caused by extreme stormwater runoff. Under these conditions, the reset gate of GRU can rapidly cut off the baseline memory formed during previous stable periods [142]. This feature gives the model an immediate and highly sensitive adaptive capacity. It enables the model to respond effectively to extreme influent fluctuations.

By continuously processing time-variant and heterogeneous operational datasets, the GRU network can capture complex UF fouling dynamics. These datasets include influent flow, localized pressure, and specific colloidal loads. Through these inputs, the model can autonomously map the relationship between complex influent matrices and subsequent non-linear TMP escalation [143]. As a result, the GRU can predict the temporal thresholds that trigger sudden resistance spikes. This prediction can be achieved with high precision [144]. This predictive capability provides critical feed-forward input for real-time hydraulic control. It allows smart systems to adaptively adjust backwash frequency and permeate flux [145]. These adjustments can be made before the system reaches critical TMP thresholds. In this way, membrane stress can be reduced. Irreversible physical compaction of the cake layer can also be prevented.

From an environmental sustainability perspective, this sequence-aware architecture changes the operational logic of UF systems. It shifts UF operation from chemically intensive reactive cleaning to physically optimized preventive control [147]. By intercepting irreversible cake-layer compaction in advance, the GRU model can greatly reduce the frequency of mandatory chemical CIP interventions [148]. This strategic shift directly reduces the discharge of highly alkaline, acidic, and chlorine-rich secondary effluents into the receiving environment [149]. Furthermore, it lowers the carbon footprint associated with chemical production. It also reduces the excessive pumping energy that is usually wasted during severe fouling events [150]. Therefore, this algorithmic advancement goes beyond computational optimization. It is closely aligned with the environmental goals of sustainable and low-impact water reclamation [151].

### 4.4. NF: Regularized XGBoost for water-quality decoupling

#### 4.4.1. Architectural logic of XGBoost

In NF systems, membrane performance is controlled by several interacting mechanisms. These include steric exclusion, hydrophobic affinities, and Donnan electrostatic shielding [152]. Datasets generated by full-scale NF facilities are often highly complex [153]. They usually show extreme high dimensionality, strong inter-variable coupling, and significant sensor background noise. A typical example is the severe multicollinearity among DOC, $UV_{254}$, pH, and divalent cation concentrations [154].

To navigate these complex matrices, the XGBoost model has become a preferred computational engine. XGBoost integrates multiple weak learners, mainly Classification and Regression Trees (CART), through an additive and incremental training framework [155]. In this framework, each newly added decision tree is mathematically designed to correct the prediction residuals left by previous models along the gradient direction [156]. The most prominent mathematical advantage of XGBoost lies in its highly rigorous regularized objective function. This function incorporates both L1 (Lasso) and L2 (Ridge) regularization penalty terms into the loss function [157]. These penalties constrain leaf node weights and overall tree depth. As a result, XGBoost effectively limits model complexity and strongly suppresses the overfitting risks commonly observed in deep learning models [158]. Furthermore, XGBoost uses a second-order Taylor expansion to approximate the loss function. This allows the model to use both first-order gradients and second-order Hessians during training. Through this mechanism, it can identify optimal and sparsity-aware feature splitting points [159].

#### 4.4.2. Alignment with NF fouling and dosing control

The dual-regularization design of XGBoost is critical for decoupling the deeply synergistic mechanisms involved in NF fouling. As illustrated in Fig. 2, XGBoost can isolate the specific mechanism by which multivalent metal cations, such as $Ca^{2+}$ and $Mg^{2+}$, cross-link with certain NOM fractions. This interaction can intensify gel layer scaling. Importantly, XGBoost can identify this mechanism without being misled by standard operational fluctuations or overlapping DOM fluorescence signatures [160]. By combining this ensemble architecture with feature importance analysis, researchers can further map complex non-linear relationships. For example, they can clarify how sustained low-molecular-weight organic loading contributes to later MWCO attenuation [161]. This analytical capability provides strong support for multi-objective optimization in NF processes. In traditional coagulation-NF hybrid systems, XGBoost can be used to continuously calculate the required chemical coagulant dosage according to dynamic raw water characteristics [162]. This targeted intervention improves the removal of problematic organic fractions in upstream treatment. It also helps preserve the reversibility of the NF fouling layer [163]. At the same time, it avoids secondary structural damage caused by residual chemicals, such as anionic polyacrylamide (APAM), accumulating on the membrane surface [164].

The integration of regularized XGBoost can provide environmental benefits when it is linked to dosing control. By improving dynamic pretreatment dosing, the model may reduce over-dosing of aluminum or iron salts and polymeric flocculants [165]. More precise dosing can also reduce heavy-metal-laden sludge. Such sludge is difficult to dispose of safely [166]. In this sense, algorithmic decoupling of complex water matrices can support green-chemistry principles. It may reduce the chemical footprint of advanced back-end municipal water treatment [167].

### 4.5. RO: CNN-GRU-SHAP for spatiotemporal diagnostics

#### 4.5.1. Need for multimodal spatiotemporal fusion

RO systems are difficult predictive environments. The main reason is that biofouling and inorganic scaling are often localized and hidden [168]. Biofilm formation usually begins as a microscopic outbreak. It can then spread under favorable hydraulic and nutrient conditions [169]. A one-dimensional time-series model, such as a standalone GRU, may therefore miss spatial information. A tabular model, such as XGBoost, may also be insufficient when image or surface data are available [170]. These models do not directly describe the morphology of localized fouling layers. Hybrid CNN-GRU-SHAP architectures address this gap. They combine spatial feature extraction, temporal modeling, and interpretability in a single workflow [171].

#### 4.5.2. Alignment with RO fouling and targeted cleaning

Within this highly complex multimodal framework, as shown on the right side of Fig. 2, the CNN acts as a non-linear extractor of high-order spatial features [172]. It can be coupled with non-invasive optical monitoring techniques, such as OCT or hyperspectral imaging. Through this coupling, the CNN uses localized multi-layer convolutional kernels to autonomously capture the morphological heterogeneity of the interfacial fouling layer [173]. In this context, the CNN functions as a “digital microscope”. It mathematically distinguishes microstructural variations from 3D biophysical morphological profiles. This allows the model to detect the early stages of bacterial cell attachment or gypsum nucleation. These early fouling events can be identified long before they affect macroscopic system hydraulics [174].

Subsequently, the dimensionally reduced spatial topological feature

maps are fused with macroscopic hydraulic time-series data. These data may include feed pressure, cross-flow velocity, and localized pH. The fused information is then fed into the downstream GRU module. Using its internal temporal gating mechanisms, the GRU dynamically constructs a predictive evolutionary model [175]. This model describes how the microscopic biofilms detected by the CNN may be further compacted under specific hydraulic stresses [176]. This spatiotemporal coupling gives the hybrid model high predictive precision when the multimodal dataset is reliable. It can accurately predict the temporal threshold at which localized microscopic scaling develops into macroscopic hydraulic failure.

#### 4.5.3. Deployment bottlenecks: Dimensionality and hardware

The integration of non-invasive optical monitoring with CNN-GRU-SHAP is promising. However, full-scale deployment still faces major challenges. The most important challenge is dimensionality [177]. High-resolution 2D image matrices generate large data volumes. When these images are fused with high-frequency 1D hydraulic series, computation increases quickly. Memory demand also rises [178]. This issue is important for water facilities with legacy control hardware. Many plants still rely on local Programmable Logic Controllers (PLC). These PLCs usually cannot support heavy tensor operations without additional edge-computing hardware [179].

To address the gap between complex multimodal fusion and limited hardware capacity, future engineering deployments must include advanced feature dimensionality reduction. They must also incorporate model compression protocols [180]. Before large optical datasets are fed into recurrent temporal layers, robust latent-space extraction techniques should be applied. For linear feature spaces, Principal Component Analysis (PCA) can be used. For non-linear feature spaces, Variational Autoencoders (VAEs) are more suitable. VAEs can decode and extract non-linear topological invariants [181]. By converting macroscopic image tensors into compressed low-dimensional latent representations, the computational burden on the downstream GRU can be greatly reduced [182].

Furthermore, dimensionality reduction should be combined with aggressive post-training model compression during edge deployment. These compression techniques include structural network pruning and precision quantization. For example, neural weights can be converted from FP32 to INT8 precision [183]. By streamlining the hybrid CNN-GRU network, low-latency models with reduced computational demands can be deployed directly on local edge computing devices within treatment facilities [184]. This edge-based deployment avoids severe cloud-communication latency. It also strengthens industrial data security [185]. More importantly, this hardware-software co-design demonstrates the practical feasibility of the CNN-GRU-SHAP architecture. It shows that the model can move beyond an idealized laboratory concept. It can become a robust and edge-deployable cognitive engine for full-scale digital twin water plants [186].

## 5. Performance benchmarking and deployment readiness

Translating theoretical computational architectures into highly regulated full-scale industrial operations requires rigorous benchmark evaluations [187]. These evaluations should examine statistical accuracy, generalization capability, and engineering robustness under severe system perturbations.

### 5.1. Accuracy, robustness and contextualized performance benchmarking

The reported predictive performance of ML models should be interpreted with caution. Very high values, such as $R^2$ values above 0.98–0.99 or RMSE reductions approaching 90%, are useful indicators of model potential, but they are not sufficient evidence of universal reliability [188]. Their engineering meaning depends strongly on dataset size, membrane process, prediction target, operating scale, validation strategy, prediction horizon, operating range and the baseline model used for comparison [189]. Therefore, this review treats these values as context-dependent literature reports rather than general performance guarantees.

For structured tabular datasets, XGBoost has shown strong performance in selected membrane-related prediction tasks, including SEC estimation, salinity prediction, permeability-loss assessment, rejection-drift evaluation and chemical-dose support [190]. In such cases, $R^2$ values close to 0.97–0.98 can be meaningful when the model is evaluated against appropriate baselines, such as linear regression, support vector regression, random forest, artificial neural networks or process-empirical models [191]. However, the same $R^2$ value may be less informative if the test set is small, randomly split from highly auto-correlated plant data, or restricted to a narrow operating range.

For UF fouling prediction, GRU models may achieve high $R^2$ values when high-frequency TMP, flux, pressure, turbidity and backwash-history data are available. This performance is most relevant when chronological splitting, rolling-origin validation, cross-season testing or independent module testing is used. Random train splitting can overestimate model reliability because adjacent SCADA observations are temporally correlated [192]. Therefore, GRU-based results should be reported together with sampling frequency, sequence length, prediction horizon, number of filtration cycles, operating days and shock-load events used for training and testing [193].

For RO or multimodal fouling diagnosis, CNN-GRU and CNN-GRU-SHAP models can substantially reduce RMSE when image, OCT, hyperspectral or spatially resolved sensor data are fused with hydraulic time series [194]. However, large RMSE reductions should be interpreted relative to the baseline model and the absolute error scale. A 90% RMSE reduction compared with a weak standalone baseline does not necessarily imply equivalent robustness under full-scale plant operation [195]. Such results should be evaluated using independent modules, different feedwater regimes, external test conditions or longer-term validation whenever possible.

To avoid overstatement, high-performance metrics should be reported together with explicit contextual information rather than being isolated in a descriptive comparison table [196]. If the original study does not report a specific item, this limitation should be stated directly rather than inferred. At a minimum, each ML-based membrane-fouling study should identify the membrane process, dominant prediction target, dataset scale, operating range, validation strategy, prediction horizon, baseline model and direct process-control relevance [15]. This reporting logic allows readers to distinguish between model accuracy obtained under narrow experimental conditions and model robustness that is likely to be transferable to plant-scale membrane operation [197].

For UF studies using GRU or other recurrent models, the reported inputs should include high-frequency TMP, flux, pressure, permeate flow, turbidity, particle count, temperature, filtration time, backwash history, air-scouring intensity and cleaning records where available. The corresponding outputs should be stated as measurable operating endpoints, such as TMP increase, normalized flux decline, permeability loss, threshold-crossing probability or time to the next backwash/CIP event [198]. These studies should also report sampling frequency, sequence length, number of filtration cycles, operating days, module scale, shock-load events and missing-data rate, and they should be validated by chronological splitting, rolling-origin testing, cross-season testing or independent module testing against baselines such as persistence models, ARIMA, SVR, random forest, LSTM or process-empirical control rules [199].

For NF studies using regularized XGBoost, the minimum contextual information should include DOC, $UV_{254}$, fluorescence indices, pH, conductivity, ionic strength, $Ca^{2+}$, $SO_4^{2-}$, alkalinity, temperature, pressure, recovery, flux, coagulant dose and pretreatment conditions. Outputs should be linked to permeability decline, rejection drift, MWCO

attenuation risk, scaling or gel-layer risk, SEC, salinity or recommended chemical-dose ranges. The dataset description should include sample number, water-quality variables, operating days, membrane type, feed matrix, laboratory/pilot/full-scale status and the measured ranges of pH, conductivity, flux and recovery [200]. Validation should use k-fold or nested cross-validation for independent tabular datasets, or chronological and rolling validation for plant data, with comparisons against linear regression, SVR, random forest, ANN and empirical dosing rules [201].

For RO or multimodal studies using CNN-GRU, CNN-GRU-SHAP or related architectures, the input space should be described through OCT images, hyperspectral images, surface or module images, pressure, differential pressure, flow, recovery, pH, conductivity, temperature, antiscalant dose, normalized salt passage and cleaning history. The prediction target should be expressed as localized biofouling or scaling risk, surface-fouling evolution, TMP or pressure-drop escalation, time to hydraulic deterioration or a module-specific warning score [202]. These studies should report the number of images or scans, spatial resolution, imaging modality, synchronized SCADA records, module or coupon scale, operating duration and image-preprocessing method. Independent module testing, time-separated validation, external feedwater testing or cross-site testing is preferable, and comparisons should include standalone CNN, standalone GRU, RF/XGBoost and rule-based alarm baselines [203]. When SHAP or another XAI method is coupled with UF, NF or RO prediction models, the same input variables, time-stamped records and quality-control flags used by the predictive model should be retained in the explanation layer [204]. The resulting driver attribution should be checked against operating records, operator knowledge, cleaning logs and known membrane chemistry before it is translated into a dose, cleaning or inspection recommendation.

Therefore, the comparison of GRU, XGBoost, CNN-GRU-SHAP and XAI-assisted models should not be interpreted as a ranking based only on $R^2$ or RMSE. It should instead be understood as a process-specific assessment of data availability, validation rigor, control relevance and deployment risk [205]. This interpretation is more consistent with full-scale membrane operation, where the practical value of an ML model depends not only on prediction accuracy but also on robustness, interpretability, operator trust and compatibility with existing SCADA and control infrastructure [206].

### 5.2. SCADA noise, data sparsity and fallback control

Reliable deployment requires explicit consideration of engineering data quality. Operational datasets from full-scale facilities are fundamentally different from datasets collected under controlled laboratory conditions [207]. In industrial SCADA systems, sensor drift, background noise, maintenance interruptions and communication failures are common [208].

In such harsh data environments, the XGBoost framework shows exceptional engineering robustness. Its built-in sparsity-aware split-finding algorithm can automatically learn optimal default paths for missing data [209]. This process occurs during the construction of decision trees. Therefore, XGBoost can reliably process SCADA logs that contain missing values and sensor disconnections. It does not need to rely heavily on complex frontend data imputation protocols [210]. This is important because such protocols may introduce bias.

In sharp contrast, sequence-aware deep learning models are highly sensitive to data integrity. This applies to both single-layer GRUs and complex CNN-GRU architectures. Missing values can directly break the temporal continuity required by recurrent gating mechanisms [211]. As a result, the network may lose its ability to retain or update historical states. To overcome this structural vulnerability, a robust frontend data imputation protocol is essential. However, current advanced imputation algorithms involve clear computational trade-offs [212]. These trade-offs are especially important when distinguishing between Missing Completely At Random (MCAR) and Missing Not At Random (MNAR) scenarios [213].

Before using computationally heavy reconstruction models, water-treatment plants should first evaluate simple estimation tools that are already familiar to process engineers [214]. Two practical options are moving-window filtering and Kalman-type state estimation. A moving median or exponentially weighted moving average (EWMA) can remove isolated spikes in TMP, flow, conductivity, pH or turbidity signals caused by electrical noise, short probe fouling or communication jitter [215]. It is most appropriate when the missing interval is short, the process is near steady state and the objective is alarm stabilization rather than reconstruction of a true hydraulic shock [216]. The window should be shorter than the intended early-warning horizon; otherwise, real rapid fouling events may be smoothed out. Kalman filtering is more suitable when a plant has a simple dynamic model for a slowly varying state, such as normalized permeability, TMP trend or permeate conductivity [217]. It combines the model-predicted state with noisy sensor observations and can provide online estimates with uncertainty. Therefore, it is useful for RO/NF monitoring under gradually changing recovery, temperature and pressure conditions [218]. However, Kalman-type methods should not be used blindly during abrupt influent shocks unless the process model and noise assumptions are recalibrated [219].

For transient MCAR scenarios, KNN interpolation offers fast execution. However, it often fails to capture abrupt and non-stationary hydraulic shock loads. Spline smoothing also has limitations. It may excessively smooth temporal sequences. This can mask critical early-warning signals of sudden physical pore blockage. For highly complex MNAR scenarios, advanced deep learning-based reconstruction methods can provide high-fidelity results [220]. Denoising Autoencoders (DAE) are one representative example. Recent progress in environmental informatics has also introduced new approaches. These include Graph Neural Networks (GNN) and Transformer-based spatiotemporal attention mechanisms. GNNs can use spatial topological relationships among distributed SCADA sensors [221]. Transformers can capture long-range temporal dependencies without relying on rigid sequential constraints [222]. Together, these methods expand the theoretical ability to handle multivariable and non-stationary hydraulic missing sequences. Nevertheless, these advanced architectures still face practical barriers. Although they perform well in theoretical MNAR reconstruction, they usually have very large computational footprints [223]. This makes low-latency edge deployment difficult. They also carry a serious risk of error propagation. For example, an imputation algorithm may hallucinate an abnormal hydraulic profile during a prolonged sensor blackout [224]. These synthetic errors may then be amplified by the downstream GRU update gates. In the worst case, this could trigger harmful false-positive CIP interventions.

A reliable predictive framework should not depend on a single algorithm. It should include a physics-informed dynamic fallback mechanism. The mechanism can use data-integrity thresholds as operational safeguards. Short signal losses may be handled by imputation when the missing rate is low and the sensor pattern remains physically plausible [225]. Severe or sustained communication failures require a different strategy. In such cases, the system should downgrade to a simpler and more robust control layer [226]. The exact thresholds should be calibrated for each plant, sensor network, and safety requirement.

Under data-sparse conditions, the framework can suspend GRU and CNN-GRU modules. This avoids unstable recurrent updates [227]. Control authority can then return to a regularized XGBoost ensemble or to conservative rule-based control [228]. XGBoost is useful because it can handle missing branches during split finding. This reduces dependence on aggressive imputation. The fallback mechanism protects the operational baseline until sensor integrity is restored [229].

## 6. Explainable and safe control strategies

In highly regulated engineering sectors, operators and process

engineers are unlikely to surrender control of critical membrane assets to black-box algorithms [230]. This is especially true in municipal drinking water safety assurance and large-scale industrial zero-liquid discharge (ZLD) systems. These algorithms must provide transparent physical and logical explanations [231]. Therefore, accurately forecasting a transmembrane pressure (TMP) surge is only a partial solution. The ultimate survival and deployment of a predictive model in full-scale industrial operations depend heavily on robust interpretability. This means that the model must translate complex mathematical vectors into clear and actionable physical or chemical intervention protocols [233].

### 6.1. Native feature importance versus SHAP

Compared with deep neural networks, tree-based ensemble models such as XGBoost have inherent advantages in interpretability [234]. During the construction of decision trees, XGBoost automatically calculates global native feature importance scores. These scores are usually based on information gain or feature frequency [235]. Such scores provide water treatment plants with macroscopic "priority lists" of fouling contributors. For example, they can identify the hydrophobic fraction of DOM as the main cause of permeability decline [236]. This interpretation may be more informative than simply attributing the decline to bulk influent turbidity [237].

However, native feature importance in tree-based models only provides globally averaged correlations. It cannot perform "instance-level" micro-tracing for localized or sudden fouling events at a specific operational moment [238]. This limitation highlights the necessity of incorporating SHAP analysis [239]. SHAP is rooted in cooperative game theory. It goes beyond basic correlation analysis. Specifically, it calculates the exact marginal contribution of each physicochemical input to a specific prediction [240]. These inputs may include precise temperature fluctuations, localized pH variations, or transient spikes in molecular weight distributions. SHAP then explains how each factor contributes to the deviation of a prediction from its baseline expectation [241]. Whether the predictive core is XGBoost or a hybrid CNN-GRU model, SHAP can serve as a high-resolution diagnostic tool. It effectively reduces the explanatory barriers associated with deep learning models [242].

### 6.2. Operator-facing XAI: From predictions to physical interventions

For membrane operators, the practical value of XAI is not the mathematical decomposition itself. Its value lies in converting a warning into an auditable reason for action [243]. In an XAI-enabled membrane dashboard, each prediction should be accompanied by four elements: the predicted risk or lead time, the dominant positive drivers, the physically plausible fouling mechanism, and the recommended bounded response [244]. SHAP is particularly useful because it explains why a specific alarm deviates from the normal baseline, rather than only reporting globally important variables [245].

At the plant scale, SHAP or related local attribution methods can support five operator-facing tasks. First, they can identify dominant fouling drivers by ranking variables such as TMP slope, differential pressure, normalized permeability, $UV_{254}$/DOC, $Ca^{2+}/Mg^{2+}$, sulfate saturation, pH deviation, recovery, antiscalant dose, temperature, ATP/EPS indicators, or OCT-derived biofilm thickness [246]. Second, they can distinguish likely fouling modes. A risk dominated by turbidity and TMP slope suggests cake-layer compaction, whereas a risk dominated by pH, calcium, sulfate, recovery, and antiscalant variables suggests scaling [13]. Third, they provide a defensible basis for cleaning decisions. A hydraulic signature may justify shorter filtration cycles or backwashing, whereas persistent chemical or biological attributions may justify chemically enhanced backwash, localized biocide dosing, or CIP [247]. Fourth, they can support chemical dosing by indicating whether risk is driven mainly by coagulant-sensitive DOM, pH-sensitive scaling, or under-dosed antiscalant. Fifth, they can warn operators about abnormal conditions [249]. If the model error is high or the SHAP pattern is inconsistent with known fouling chemistry, the system should flag possible sensor drift, data loss, unusual influent quality, or actuator malfunction instead of issuing an automatic command [250].

For example, a hybrid RO model may predict that TMP or differential pressure will exceed an alarm threshold within the next 48 h. A conventional black-box alarm would only tell the operator that fouling risk is high. A SHAP-enhanced alarm can indicate whether the main contribution comes from localized biofilm thickness, rising pressure drop, pH drift away from the antiscalant chelation range, increased recovery, or elevated scaling tendency [251]. The recommended response can therefore be more selective: pH correction and antiscalant adjustment for scaling-driven risk; localized biocide dosing and module inspection for biofouling-driven risk; brine flushing or recovery reduction for supersaturation-driven risk; and full-line CIP only when repeated warnings and poor post-cleaning recovery indicate irreversible deterioration [252].

The architecture in Fig. 4 should therefore be interpreted as an operator-supervised diagnostic loop rather than a fully unconstrained autonomous controller. The CNN-GRU module provides an early warning, SHAP converts the warning into ranked physical or chemical drivers, and the decision-support layer maps these drivers to bounded interventions. Post-intervention data, such as TMP recovery, normalized flux, salt passage, chemical consumption, and module pressure drop, are then returned to the model for feedback calibration.

Importantly, XAI outputs should be treated as decision support rather than proof of causality [253]. A high SHAP value for pH, $Ca^{2+}$, or OCT-derived biofilm thickness should be checked against plant observations, water chemistry, cleaning history, and manufacturer limits before it is converted into dosing or cleaning action [254]. This human-in-the-loop design allows XAI to improve operator trust while avoiding unsafe over-reaction to a single model prediction [255].

In practical use, XAI outputs should be converted into bounded operating responses according to the dominant physical or chemical driver [256]. When local attribution is dominated by TMP slope, turbidity or flux decline, the warning is most consistent with rapid cake-layer formation or compaction; the corresponding response is to shorten the filtration cycle, increase backwashing or reduce flux while confirming that TMP/flux limits and hydraulic recovery remain acceptable [257]. When $UV_{254}$, DOC, divalent-cation indicators or pH dominate the explanation, the likely concern is organic-inorganic fouling or charge shielding, so coagulant dose, pH and pretreatment settings should be adjusted only within approved chemical-dose ranges [258].

For RO operation, attributions dominated by pH, recovery, sulfate saturation or antiscalant dose point to localized scaling, for which pH correction, antiscalant adjustment, recovery reduction or brine-channel flushing should be checked against saturation indices and membrane limits [259]. By contrast, OCT-derived biofilm thickness, ATP/EPS indicators or pressure-drop increase indicate localized biofouling risk and should trigger module inspection, targeted cleaning or localized biocide dosing after confirmation by pressure-drop trends or microbial indicators [260]. High residual error or inconsistent attribution patterns should be treated as possible sensor drift, data loss or abnormal influent disturbance; alarm confirmation, conservative fallback control and human review should therefore precede automatic actuation [261].

### 6.3. Critical barriers to trustworthy ML-based membrane control

Several barriers still limit the direct translation of ML predictions into autonomous membrane control. First, many published studies rely on random train-test splits from a single experimental campaign or one treatment facility [262]. This practice can overestimate predictive performance because adjacent time points often share the same hydraulic history, influent regime and cleaning cycle. Future studies should report chronological validation, cross-season testing and, where possible, cross-site validation.

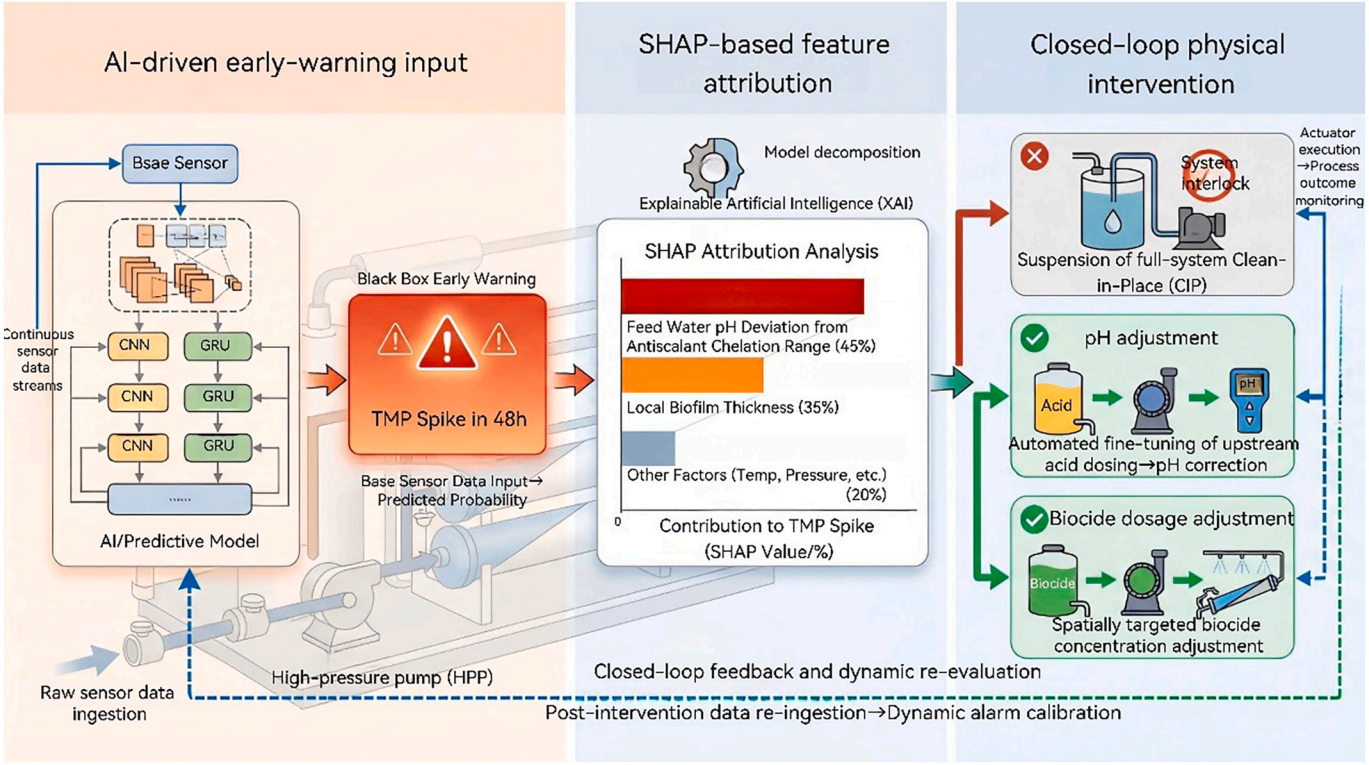



**Fig. 4.** Conceptual framework of an XAI-driven closed-loop autonomous control system in membrane facilities.

Second, high statistical accuracy does not automatically imply safe control [263]. A model that predicts TMP accurately may still recommend an unsafe operating action if the control boundary is not constrained by membrane integrity, allowable recovery, cleaning chemistry, regulatory limits or operator practice. Therefore, ML outputs should be embedded in supervisory control layers that include hard safety limits, alarm confirmation and human override [264].

Third, explainability methods should be interpreted carefully. SHAP values can identify variables that contribute strongly to a model prediction, but they do not prove mechanistic causality by themselves [265]. In membrane systems, attribution results should be checked against known fouling chemistry, mass-transfer limits and plant observations before they are converted into chemical dosing or cleaning decisions [266].

Finally, open and standardized datasets remain scarce in the membrane sector [267]. This limits fair comparison among GRU, XGBoost, CNN-GRU and emerging Transformer or graph-based models. Future benchmarking should report sensor lists, sampling frequency, missing-data rate, cleaning events, validation design, uncertainty metrics and deployment constraints [268]. These details are essential for assessing whether a model is only accurate in one dataset or transferable to real water-process operation [269].

## 7. Comparison with other reviews

To clarify the novelty and boundary of the present review, Table 2 provides a concise comparison with representative previous reviews on membrane fouling, AI/ML-assisted membrane operation and process-specific fouling control. The comparison focuses on review scope, membrane processes, fouling mechanisms, ML models, process-control

**Table 2**
Comparison between the present review and representative previous reviews on membrane fouling and AI/ML-assisted membrane processes.

| Review group | Scope / processes | Fouling mechanisms | AI/ML and XAI | Control / DT | Gap filled by this review | Ref. |
|---|---|---|---|---|---|---|
| Broad membrane-technology reviews | Broad membrane technologies | General fouling and antifouling | Limited AI; XAI not central | Control and digital twin not central | No mechanism-model-control link | [270] |
| Dynamic fouling-characterization reviews | Pressure-driven fouling dynamics | Cake, biofilm and scaling evolution | Model selection not central | Monitoring-focused | No action-oriented ML translation | [271] |
| Antifouling and mitigation reviews | Materials and mitigation strategies | Pore blocking, organic fouling, biofouling and scaling | Limited AI/XAI | Mitigation-focused; limited digital-twin linkage | Limited plant-data deployment logic | [272] |
| AI/ML fouling-prediction reviews | AI/ML prediction and MBR operation | Fouling mainly treated as prediction target | ANN, SVM, RF and related models; limited XAI | Accuracy-focused; bounded control less developed | Weak link to plant setpoints | [273] |
| RO/NF-specific fouling reviews | NF/RO fouling and control | Scaling, biofouling, cleaning and rejection loss | Process-specific AI partly discussed | Control within NF/RO; limited digital-twin coverage | Limited UF-NF-RO transferability | [274] |
| Present review | UF, NF and RO water-treatment membranes | UF cake; NF DOM-cation coupling; RO biofilm/scaling | GRU, XGBoost and CNN-GRU-SHAP; XAI included | Backwash, dosing, pH, antiscalant, CIP and edge/digital twin | Mechanism-data-model-control framework | This review |

relevance, XAI, digital-twin coverage and the specific gap addressed here.

As summarized in Table 2, previous reviews provide important foundations for membrane technology development, fouling characterization, anti-fouling strategies and AI-assisted prediction. However, they usually emphasize broad applications, a single process level or algorithmic accuracy. In contrast, the present review adds a mechanism-to-control perspective by linking UF, NF and RO fouling pathways to data signatures, model architectures and bounded operator actions. This positioning makes the contribution more explicit: GRU is positioned for rapid UF hydraulic sequences, regularized XGBoost for multicollinear NF water-quality interactions, and CNN-GRU-SHAP for spatially heterogeneous RO biofouling/scaling, with XAI and digital-twin deployment treated as requirements for safe plant-scale use.

## 8. Future perspectives

Future research should move beyond laboratory-scale model demonstrations toward validated, safe and transferable control systems for full-scale membrane facilities. Four directions are particularly important: physics-informed learning, plant-wide digital twins, privacy-preserving model training and open benchmarking with standardized fouling metrics and techno-economic assessment.

As illustrated in Fig. 5, the future development of smart membrane-based water treatment can be understood as a staged evolution from data-driven monitoring to physics-informed autonomous operation. In the first phase, physical water treatment plants are equipped with sensors, instrumentation, actuators and chemical dosing units. These devices generate real-time hydraulic, water-quality and operational data. In the second phase, these data streams are processed by edge-computing nodes and physics-informed neural networks. The models no longer depend only on statistical correlations. Instead, they are constrained by physical laws, such as fluid dynamics, mass transfer and membrane transport equations. In the third phase, the physical plant is connected to a virtual water treatment plant. This digital-twin environment allows cleaning protocols, dosing strategies and hydraulic adjustments to be tested in a zero-physical-risk simulation space before they are applied to the real system. Therefore, Fig. 5 extends the central logic of this review. It shows how mechanism-aligned ML can evolve from fouling prediction to safe process control and finally to autonomous optimization in next-generation water treatment facilities.

As shown in Fig. 5, the roadmap shows the transition from data-driven physical plants to physics-informed edge computing and autonomy supported by digital twins. Sensor data, edge-computed information and control commands are integrated with PINN-based physical constraints. This integration supports virtual testing of cleaning, dosing and backwash strategies before plant-level implementation.

### 8.1. Thermodynamic robustness through PINNs

Future ML frameworks should move beyond purely data-driven correlations. Physics-Informed Machine Learning (PIML) provides one important pathway. Physics-Informed Neural Networks (PINNs) can embed governing equations into the loss function. These equations may include Navier–Stokes flow, concentration polarization, Donnan steric transport or membrane mass-transfer terms. This design can constrain model predictions with mass conservation, energy conservation and transport physics. It may reduce the risk of physically invalid control recommendations under hydraulic shocks, influent disturbances or SCADA noise.

For membrane-fouling prediction, three governing equations are particularly suitable for PINN embedding, but they should be written in explicit vector form so that the residual terms remain physically meaningful.

$$\nabla u = 0 \; and \; \rho\left(\frac{\partial u}{\partial t} + u\nabla u\right) = -\nabla p + \mu\nabla^2 u \tag{1}$$

This equation represents incompressible continuity and momentum conservation in the feed channel. In a PINN, its residual can constrain cross-flow velocity, pressure drop and shear stress, and it is most useful when local hydrodynamics controls particle deposition, cake compression or biofilm detachment. Here, $\mu$ is the velocity vector, $\rho$ is fluid density, $t$ is time, $p$ is pressure, $\mu$ is dynamic viscosity, $\nabla$ is the gradient/

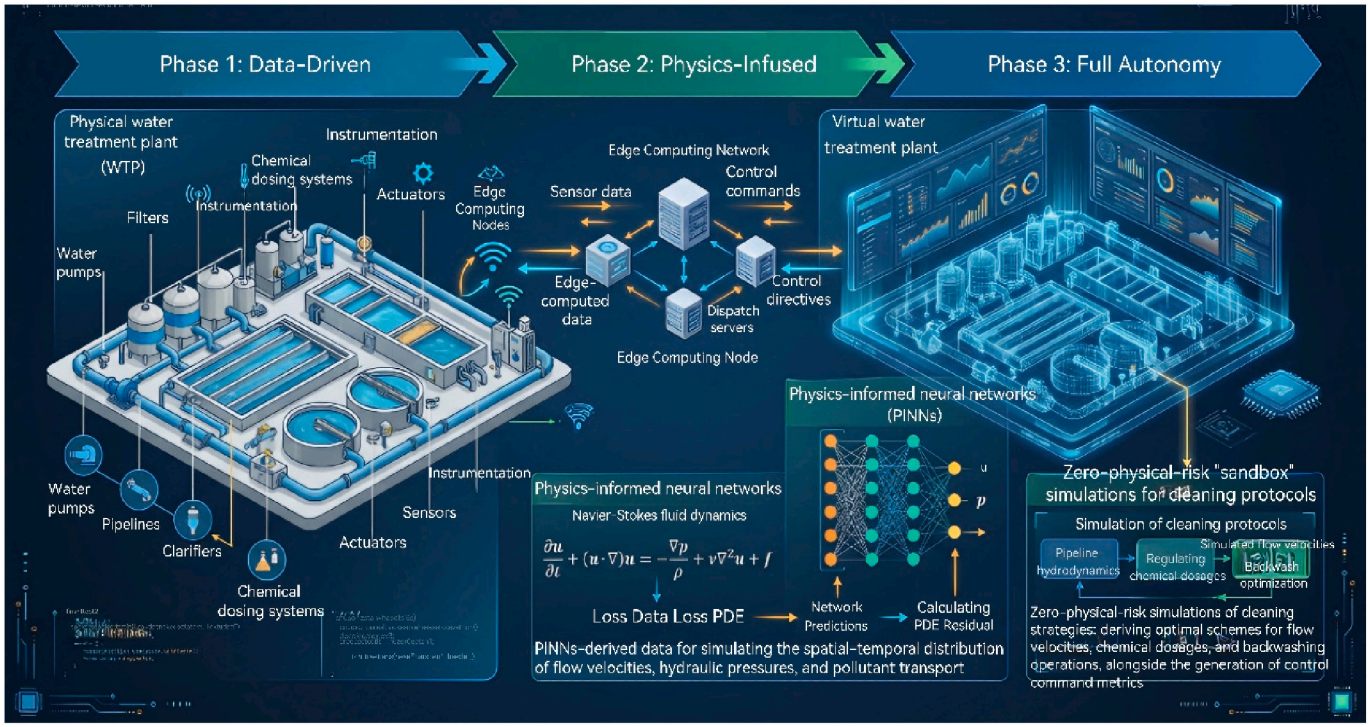



**Fig. 5.** Evolutionary roadmap for next-generation smart water treatment facilities.

divergence operator and $\nabla^2$ is the Laplacian operator.

$$\frac{\partial C}{\partial t} + u\nabla C = \nabla(D\nabla C) - R(C) \tag{2}$$

This equation represents advection-diffusion-reaction mass balance. It can constrain concentration polarization, solute accumulation and scaling-precursor transport near the membrane surface, especially in NF and RO systems where $Ca^{2+}$, $SO_4{}^{2-}$, DOM or nutrients accumulate at the interface. Here, $C$ is solute or foulant concentration, $D$ is the effective diffusion coefficient, and $R(C)$ is the net reaction, precipitation, biological-growth or removal term. When $D$ is constant, $\nabla(D\nabla C)$ can be simplified to $D\nabla^2 C$.

$$Jv = \frac{\Delta P - \Delta\pi}{[\mu(Rm + Rf)\,]\ and\ Rf}, \quad Rf = Rc + Rp + Rb \tag{3}$$

This equation represents Darcy's law combined with a resistance-in-series expression and an osmotic-pressure correction. It can constrain the relationship among water flux, TMP, osmotic pressure, viscosity and fouling resistance, making it suitable for predicting TMP rise, permeability loss and cleaning thresholds in UF, NF and RO operation. Here, $Jv$ is permeate flux, $\Delta p$ is the applied transmembrane pressure, $\Delta\pi$ is the osmotic pressure difference across the membrane, $Rm$ is intrinsic membrane resistance, $Rf$ is total fouling resistance, and $Rc$, $Rp$ and $Rb$ denote cake/gel-layer resistance, pore-blocking or pore-constriction resistance and biofilm-related resistance, respectively. For UF, $\Delta\pi$is often small and may be neglected, whereas for NF and RO it should be retained to avoid thermodynamically unrealistic flux predictions.

In practice, the PINN loss function can combine the data-fitting error with weighted residual penalties from one or more of these equations. The selected equation should match the available sensors and the target control action, rather than being inserted only to increase model complexity.

This direction is particularly important for membrane operation because fouling is not only a statistical pattern. It is also governed by physical deposition, chemical interaction, biological growth and hydraulic stress. For UF systems, PINNs may help constrain GRU-based TMP prediction within realistic hydraulic boundaries. For NF systems, physics-informed constraints may improve the interpretation of organic–inorganic coupling and Donnan shielding. For RO systems, PINNs may support safer prediction of localized scaling and biofilm growth under changing recovery, pH and antiscalant conditions. In this way, physics-informed learning can strengthen the mechanistic reliability of the ML architectures discussed in this review.

### 8.2. Plant-wide digital twins for multi-barrier optimization

Current predictive modeling often focuses on individual membrane modules. Future work should scale these models toward plant-wide digital twins. This is important for multi-barrier treatment trains, such as $O_3$-coagulation-UF-NF systems, RO-based wastewater reuse lines and decentralized drinking-water networks. In these systems, fouling risk does not develop in a single unit alone. Upstream coagulation, oxidation, filtration and chemical dosing can strongly influence downstream membrane resistance.

As shown in Fig. 5, a digital twin can connect the physical water treatment plant with a virtual plant. The virtual plant can simulate hydraulic conditions, cleaning protocols, chemical dosing and backwash optimization before real-world implementation. This provides a safer control environment than direct trial-and-error operation. Edge computing can help synchronize virtual simulations with physical plant data. Multimodal models can then serve as cognitive engines for early warning, process diagnosis and control optimization. The goal is to simulate cross-stage fouling propagation before failure occurs. Such plant-wide digital twins could help operators shift from reactive membrane cleaning to predictive and preventive process management.

### 8.3. Federated learning to overcome data silos

Advanced deep neural networks can provide strong diagnostic precision. However, their generalization is still limited by data silos in the water sector. Many utilities cannot share high-resolution SCADA records because of infrastructure security, commercial confidentiality and data-governance constraints. This limitation makes it difficult to train robust models across different water matrices, membrane materials, operating modes and climatic conditions.

Federated learning offers a possible solution. It allows models to be trained at local edge nodes while only protected model updates are exchanged among facilities. Raw operational data can remain within each plant. This approach may support more generalizable AI models while protecting sensitive plant information. In the context of Fig. 5, federated learning can extend the digital twin concept from a single plant to a distributed network of smart water treatment facilities. Each facility can contribute to shared model improvement without exposing its raw SCADA data. This strategy may be especially useful for validating GRU, XGBoost, CNN-GRU-SHAP and PINN-based models across different sites.

Despite these opportunities, federated learning should not be treated as a complete solution by itself. Differences in sensor configuration, sampling frequency, cleaning records, membrane type and influent composition can still cause model bias. Therefore, future studies should combine federated learning with standardized data dictionaries, uncertainty quantification, cross-site validation and operator-supervised control boundaries. Only through this combined strategy can ML-based membrane control become both transferable and trustworthy.

### 8.4. Open benchmarks, standardized fouling metrics and techno-economic assessment

To make these future directions operational, the field needs shared benchmarks, standardized fouling indicators and techno-economic evaluation rules that are explicitly linked to plant operation. Open UF, NF and RO datasets should report the process type, feed matrix, membrane and module type, SCADA variables, water-quality variables, sampling frequency, missing-data rate, pretreatment conditions, chemical-dosing records, cleaning events, abnormal-operation labels and independent test periods. For time-series data, chronological, seasonal and cross-site test sets should be provided. For image-based RO or NF studies, OCT, hyperspectral or surface-imaging data should be synchronized with hydraulic and chemical variables. This structure would allow GRU, XGBoost, CNN-GRU-SHAP, PINN and future Transformer or graph-based models to be compared under the same engineering conditions.

Standardized fouling and control metrics should also be reported to connect model accuracy with operator-readable membrane status. Recommended indicators include normalized permeability, TMP slope, flux decline, pressure drop, salt passage, rejection drift, cleaning frequency, recovery after cleaning, chemical dose and SEC. Model validation should move beyond random data splitting and include chronological splitting, rolling-origin validation, seasonal testing, cross-module testing, cross-site testing and uncertainty intervals. These requirements would reduce over-optimistic accuracy claims and test whether a model remains transferable under different influent matrices, membrane modules and operating histories.

Finally, future plant-scale studies should evaluate baseline operation against AI-assisted operation over sufficiently long periods and quantify energy use, chemical consumption, CIP frequency, membrane lifetime, water recovery, downtime, sludge production, greenhouse-gas implications and the risk of off-specification permeate. Techno-economic reporting should include sensor and edge-hardware CAPEX, energy, chemical, labor, membrane-replacement and downtime OPEX, payback period and sensitivity analysis. Before autonomous control is attempted, studies should also define control-readiness conditions, including safety

limits, actuator constraints, alarm confirmation, human override, cybersecurity requirements and fallback-control logic. Together, these reporting elements would convert model outputs into bounded and auditable operating actions and make plant-scale validation, physics-informed learning, digital twins and federated learning easier to judge by utilities, regulators and technology providers.

From an engineering implementation perspective, water utilities should first establish a reliable data foundation by auditing SCADA tags, calibrating key sensors, standardizing fouling indicators and linking operating data with cleaning and dosing logs. They should then deploy AI models in an advisory mode, where predictions are benchmarked against operator decisions and translated into bounded recommendations for flux, backwash, pH, antiscalant or coagulant adjustment under explicit safety constraints. After sufficient plant-scale validation, these advisory models can be connected to model predictive control and digital twins so that utilities can gradually shift from alarm-driven reactive cleaning to AI-assisted predictive control with human override and fallback protection.

## 9. Conclusions

Membrane fouling remains a major barrier to stable, low-carbon and chemically efficient membrane operation. This review shows that model selection should be driven by fouling mechanism, data structure and control objective rather than by algorithmic complexity alone. UF systems require models that can follow rapid cake-layer development and hydraulic memory. GRU models are suitable when high-frequency time-series data are reliable and when the output is linked to adaptive backwashing or flux control. NF systems require models that can handle coupled organic, inorganic and electrostatic variables. Regularized XGBoost is useful because it can decouple high-dimensional tabular features and tolerate missing values. It is especially valuable when the output supports pretreatment or dosing decisions. RO systems require models that can capture spatially localized biofouling and scaling. Hybrid CNN-GRU models are suitable when optical or image-based data are available and when the results are linked to targeted cleaning, pH correction or antiscalant control.

This review further emphasizes that high prediction accuracy alone is not sufficient for engineering deployment. Predictive models must be interpretable, robust to SCADA noise and compatible with plant-level safety constraints. XAI methods, especially SHAP, are needed to translate model outputs into operator-readable physical or chemical interventions. Fallback control is also necessary when data integrity is weakened by sensor drift, signal loss or communication failures. Overall, future ML-based fouling control should be evaluated by deployability rather than prediction accuracy alone. A useful framework should provide reproducible benchmarks, use standardized fouling indicators and demonstrate economic and environmental value under plant-relevant constraints.

## CRediT authorship contribution statement

**Xin Yang:** Writing – original draft, Investigation. **Jundong Zhao:** Methodology. **Junzhao Zhuang:** Project administration. **Songlin Guan:** Writing – review & editing. **Chunbo Wang:** Validation. **Kenglin Huang:** Software. **Lele Liu:** Data curation. **Yunan Gao:** Supervision, Formal analysis.

## Funding

The authors are thankful to Foshan Self-funded Science & Technology Innovation Project (2420001003512); GuangDong Basic and Applied Basic Research Foundation of Guangdong Province (2022A1515140184).

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

The authors are thankful to Foshan Self-funded Science & Technology Innovation Project (2420001003512); GuangDong Basic and Applied Basic Research Foundation of Guangdong Province (2022A1515140184).

## Data availability

No data was used for the research described in the article.

## References

[1] S. Foorginezhad, M.M. Zerafat, A.F. Ismail, P.S. Goh, Emerging membrane technologies for sustainable water treatment: a review on recent advances, Environ. Sci.: Adv. 4 (2025) 530–570, https://doi.org/10.1039/D4VA00378K.
[2] Y.-R. Zhang, T.-W. Wu, I.-L. Chien, Intensified hybrid reactive-extractive distillation process for the separation of water-containing ternary mixtures, Sep. Purif. Technol. 279 (2021) 119712, https://doi.org/10.1016/j.seppur.2021.119712.
[3] S. Kumar Singh, C. Sharma, A. Maiti, Forward osmosis to treat effluent of pulp and paper industry using urea draw-solute: Energy consumption, water flux, and solute flux, Sep. Purif. Technol. 278 (2021) 119617, https://doi.org/10.1016/j.seppur.2021.119617.
[4] S. Liang, K. Fu, X. Li, Z. Wang, Unveiling the spatiotemporal dynamics of membrane fouling: A focused review on dynamic fouling characterization techniques and future perspectives, Adv. Colloid Interf. Sci. 328 (2024) 103179, https://doi.org/10.1016/j.cis.2024.103179.
[5] J. Liang, K. Gao, A. Zhou, Y. Fang, S. Su, L. Fu, M. Sun, X. Duan, Long-lasting performance of high-flux perovskite membrane for catalytic degradation of organic pollutants, Appl. Catal. B Environ. 327 (2023) 122440, https://doi.org/10.1016/j.apcatb.2023.122440.
[6] Y.-H. Cheng, Membrane fouling study : from fundamental theory to industrial fouling mitigation. https://hdl.handle.net/2152/86942, 2021 (accessed May 18, 2026).
[7] I.G. Wenten, M.Z. Bazant, K. Khoiruddin, Mitigating electrodialysis membrane fouling in seawater desalination, Sep. Purif. Technol. 345 (2024) 127228, https://doi.org/10.1016/j.seppur.2024.127228.
[8] Y. Xie, Q. Zhang, Q. Wu, J. Zhang, M. Dzakpasu, X.C. Wang, A modified anaerobic-anoxic-oxic system for optimizing rural sewage treatment with fluctuating influent flows, Chem. Eng. J. 481 (2024) 148492, https://doi.org/10.1016/j.cej.2023.148492.
[9] J. Song, Q. Sun, A.G. Skirtach, B.V. Parakhonskiy, Machine learning in nanoscience and beyond: Workflows, data processing, XAI and ITAP metrics, language-based models, Adv. Colloid Interf. Sci. 349 (2026) 103732, https://doi.org/10.1016/j.cis.2025.103732.
[10] M. Fei, Z. Zhang, W. Zhao, P. Zhang, Z. Xing, Optimal power distribution control in modular power architecture using hydraulic free piston engines, Appl. Energy 358 (2024) 122540, https://doi.org/10.1016/j.apenergy.2023.122540.
[11] H. Zhang, L. Shi, J. Cai, X. Wang, Z. Yan, DMGTNet: A graph-temporal deep learning model for membrane fouling prediction in dual-membrane wastewater treatment systems, J. Membr. Sci. 736 (2025) 124648, https://doi.org/10.1016/j.memsci.2025.124648.
[12] Y. Liu, S. Gong, M. Zhang, X. Zhang, H. Wang, H. Wen, S. Chang, W. Guo, H. Hao Ngo, Enhanced anti-fouling performance through integrated coagulation and membrane co-deposition modification for sustainable water treatment, Sep. Purif. Technol. 361 (2025) 131407, https://doi.org/10.1016/j.seppur.2025.131407.
[13] C. Chen, Y. Yang, N.J.D. Graham, Z. Li, X. Yang, Z. Wang, N. Farhat, J. S. Vrouwenvelder, L. Hou, A comprehensive evaluation of the temporal and spatial fouling characteristics of RO membranes in a full-scale seawater desalination plant, Water Res. 249 (2024) 120914, https://doi.org/10.1016/j.watres.2023.120914.
[14] L. Wang, Z. Li, J. Fan, Z. Han, The intelligent prediction of membrane fouling during membrane filtration by mathematical models and artificial intelligence models, Chemosphere 349 (2024) 141031, https://doi.org/10.1016/j.chemosphere.2023.141031.
[15] Y. Zhu, Y. Wang, E. Zhu, Z. Ma, H. Wang, C. Chen, J. Guan, T.D. Waite, Predicting Membrane Fouling of Submerged Membrane Bioreactor Wastewater Treatment Plants Using Machine Learning, Environ. Sci. Technol. 59 (2025) 10010–10021, https://doi.org/10.1021/acs.est.4c12835.
[16] B.G. Akpan, Conceptual model for integrating nanotechnology and artificial intelligence in sustainable water quality management, NIPES - J. Energy Technol. Environ. 7 (2025) 252–269, https://doi.org/10.37933/jete/7.4.2025.2575.

[17] N. Zareei, H. Qiblawey, Seasonal variations in feed-water chemistry and fouling dynamics of reverse-osmosis systems: A global climate lens, Desalination 622 (2026) 119744, https://doi.org/10.1016/j.desal.2025.119744.
[18] I. Ćurić, D. Dolar, J. Bošnjak, Reuse of textile wastewater for dyeing cotton knitted fabric with hybrid treatment: Coagulation/sand filtration/UF/NF-RO, J. Environ. Manag. 295 (2021) 113133, https://doi.org/10.1016/j.jenvman.2021.113133.
[19] S.M. Abdullah, M. Periyasamy, N.A. Kamaludeen, S.K. Towfek, R. Marappan, S. Kidambi Raju, A.H. Alharbi, D.S. Khafaga, Optimizing traffic flow in smart cities: soft GRU-based recurrent neural networks for enhanced congestion prediction using deep learning, Sustainability 15 (2023) 5949, https://doi.org/10.3390/su15075949.
[20] A. Hu, Y. Liu, X. Wang, S. Xia, B. Van der Bruggen, A machine learning based framework to tailor properties of nanofiltration and reverse osmosis membranes for targeted removal of organic micropollutants, Water Res. 268 (2025) 122677, https://doi.org/10.1016/j.watres.2024.122677.
[21] M.G. Zamani, M.R. Nikoo, G. Al-Rawas, R. Nazari, D. Rastad, A.H. Gandomi, Hybrid WT–CNN–GRU-based model for the estimation of reservoir water quality variables considering spatio-temporal features, J. Environ. Manag. 358 (2024) 120756, https://doi.org/10.1016/j.jenvman.2024.120756.
[22] Y. Shen, A.R. Badireddy, A critical review on electric field-assisted membrane processes: implications for fouling control, water recovery, and future prospects, Membranes 11 (2021) 820, https://doi.org/10.3390/membranes11110820.
[23] S.H. Godasiaei, Exploring the influence of crystallization fouling on microscale heat exchangers through machine learning analysis, Numer. Heat Transf. A Appl. 86 (2025) 8022–8048, https://doi.org/10.1080/10407782.2024.2357582.
[24] L. Yang, C. Xia, J. Jiang, X. Chen, Y. Zhou, C. Yuan, L. Bai, S. Meng, G. Cao, Removal of antibiotics and estrogens by nanofiltration and reverse osmosis membranes, J. Hazard. Mater. 461 (2024) 132628, https://doi.org/10.1016/j.jhazmat.2023.132628.
[25] I. Chimanlal, L.N. Nthunya, O.T. Mahlangu, B. Kirkebæk, A. Ali, C.A. Quist-Jensen, H. Richards, Nanoparticle-enhanced PVDF flat-sheet membranes for seawater desalination in direct contact membrane distillation, Membranes 13 (2023) 317, https://doi.org/10.3390/membranes13030317.
[26] M. Su, Y. Zhang, S. Liu, Y. Wang, T. Li, Challenges and solutions for nanofiltration membranes in water treatment, Front. Chem. Eng. 7 (2025), https://doi.org/10.3389/fceng.2025.1695014.
[27] S. Mazinani, A. Al-Shimmery, Y.M.J. Chew, D. Mattia, 3D printed nanofiltration composite membranes with reduced concentration polarisation, J. Membr. Sci. 644 (2022) 120137, https://doi.org/10.1016/j.memsci.2021.120137.
[28] Y. Maeda, Fouling of reverse osmosis (RO) and nanofiltration (NF) membranes by low molecular weight organic compounds (LMWOCs), part 2: countermeasures and applications, Membranes 15 (2025) 94, https://doi.org/10.3390/membranes15030094.
[29] S.K. Lau, W.F. Yong, Recent Progress of Zwitterionic Materials as Antifouling Membranes for Ultrafiltration, Nanofiltration, and Reverse Osmosis, ACS Appl. Polym. Mater. 3 (2021) 4390–4412, https://doi.org/10.1021/acsapm.1c00779.
[30] B.A. Pak, Y. Bal, E.K. Demir, B. Ahmed Hatib, B. Güzel, O. Canlı, E.E. Öktem, A. Bayrakdar, E. Sahinkaya, Micropollutant removal from agricultural drainage waters by nanofiltration and reverse osmosis processes, J. Chem. Technol. Biotechnol. n/a (n.d.). doi:https://doi.org/10.1002/jctb.70219.
[31] E. Alonso, C. Sanchez-Huerta, Z. Ali, Y. Wang, L. Fortunato, I. Pinnau, Evaluation of nanofiltration and reverse osmosis membranes for efficient rejection of organic micropollutants, J. Membr. Sci. 693 (2024) 122357, https://doi.org/10.1016/j.memsci.2023.122357.
[32] J. Wentrup, I. Bösing, T. Dülcks, J. Thöming, Rapid online analysis of n-alkanes in gaseous streams via APCI mass spectrometry, Anal. Bioanal. Chem. 416 (2024) 1843–1855, https://doi.org/10.1007/s00216-024-05182-3.
[33] J. Lakner, G. Lakner, Prediction of expected fouling time during transmembrane transition in reverse osmosis systems, Membranes 15 (2025) 170, https://doi.org/10.3390/membranes15060170.
[34] O.T. Mahlangu, L.N. Nthunya, M.M. Motsa, E. Morifi, H. Richards, B.B. Mamba, Fouling of high pressure-driven NF and RO membranes in desalination processes: Mechanisms and implications on salt rejection, Chem. Eng. Res. Des. 199 (2023) 268–295, https://doi.org/10.1016/j.cherd.2023.09.037.
[35] X. Tong, Z.-W. Zhang, Y.-H. Wu, Y. Bai, N. Ikuno, K. Ishii, H.-Y. Hu, Ultrafiltration significantly increased the scaling potential of municipal secondary effluent on reverse osmosis membranes, Water Res. 220 (2022) 118672, https://doi.org/10.1016/j.watres.2022.118672.
[36] X. Ge, Q.-L. Han, X.-M. Zhang, D. Ding, Dynamic Event-triggered Control and Estimation: A Survey, Int. J. Autom. Comput. 18 (2021) 857–886, https://doi.org/10.1007/s11633-021-1306-z.
[37] J. Kucera, Reverse Osmosis, John Wiley & Sons, 2023.
[38] O. Surinta, Effective data augmentation and training techniques for improving deep learning in plant leaf disease recognition, Appl. Sci. Eng. Prog. (2021), https://doi.org/10.14416/J.ASEP.2021.01.003.
[39] W.B. Yates, E.C. Keedwell, Offline Learning with a Selection Hyper-Heuristic: An Application to Water Distribution Network Optimisation, Evol. Comput. 29 (2021) 187–210, https://doi.org/10.1162/evco_a_00277.
[41] H. Qadi, A Study of the Opportunities and Challenges of Using Edge Computing to Accelerate Cloud Applications. https://hdl.handle.net/10012/21476, 2025 (accessed June 21, 2026).
[43] H. Wang, Y. Zhang, C. Lin, A. Wang, Y. Wang, Y. Hu, S. Jiang, J. Wang, H. Liang, X. Tang, Effects of pre-chlorination on ultrafiltration process in directly treating seasonal high-turbidity surface water: Membrane fouling control and shock load resisting, J. Water Process Eng. 68 (2024) 106356, https://doi.org/10.1016/j.jwpe.2024.106356.
[44] H.-Y. Guo, X.-M. Wang, K. Wang, S. Liu, Adsorption of natural organic matter and divalent cations onto / inside loose nanofiltration membranes: Implications for drinking water treatment from rejection selectivity perspective, Water Res. 282 (2025) 123660, https://doi.org/10.1016/j.watres.2025.123660.
[45] X. Wang, C. Ruan, C. Shen, J. Liao, D. Wang, P.J.J. Alvarez, P. Yu, Synergistic Treatment of Reverse Osmosis Membrane Biofouling with Quorum Quenching Bacteria and Hitchhiking Phages, Environ. Sci. Technol. 59 (2025) 5647–5660, https://doi.org/10.1021/acs.est.4c12852.
[46] D. AbuKhadra, B. Poodiack, L. Ghazaryan, T. Coves, Y. Oren, O. Gillor, M. Herzberg, Revisiting the role of EPS in ultrafiltration membrane fouling via LSPR-QCMD and biofilm analysis, Desalination 629 (2026) 120150, https://doi.org/10.1016/j.desal.2026.120150.
[47] Z. Wang, X. You, C. Yang, W. Li, Y. Li, Y. Li, J. Shen, R. Zhang, Y. Su, Z. Jiang, Ultrathin polyamide nanofiltration membranes with tunable chargeability for multivalent cation removal, J. Membr. Sci. 642 (2022) 119971, https://doi.org/10.1016/j.memsci.2021.119971.
[49] T. Zheng, S. Wen, Y. Wu, P. Shuai, D. Hou, Y. Jin, Membrane fouling mechanisms in the microfiltration of oat protein–β-glucan complexes, Membranes 16 (2026) 116, https://doi.org/10.3390/membranes16040116.
[50] X. Chi, K. Liu, L. Yu, N. Li, Effect of different sinusoidal dynamic water pressures on the anticlogging performance of labyrinth emitters, J. Irrig. Drain. Eng. 152 (2026) 04026004, https://doi.org/10.1061/JIDEDH.IRENG-10707.
[51] Y. Wang, H. Xu, H. Yao, B. Liu, M. Ding, T. Lin, T. Mo, L. Gao, L. Zhang, Insights into the role of prechlorination in algae-laden raw water distribution process: Algal organic matter and microcystin-LR release, extracellular polymeric substances (EPS) aggregation, and pipeline biofilm communities, J. Hazard. Mater. 443 (2023) 130306, https://doi.org/10.1016/j.jhazmat.2022.130306.
[52] M. Wei, Y. Zhang, X. Zheng, C. Zhang, J. Duan, K. Guo, M. Wang, D. Li, T. Huang, R. Chen, Residual polyacrylamide-induced membrane fouling in short-flow coagulation-ultrafiltration: Behavior and mechanism, Chem. Eng. J. 521 (2025) 166353, https://doi.org/10.1016/j.cej.2025.166353.
[53] D. Li, Z. Wang, Q. Ding, H. Sun, S. Fang, K. Zhang, W. Hu, J. Bian, Interactions between dissolved organic matter with different molecular weights and nonylphenol in surface water bodies, Chemosphere 367 (2024) 143592, https://doi.org/10.1016/j.chemosphere.2024.143592.
[54] C. Li, W. Sun, Z. Lu, X. Ao, S. Li, Z. Wang, F. Qi, O. Ismailova, Contribution of filtration and photocatalysis to DOM removal and fouling mechanism during in-situ UV-LED photocatalytic ceramic membrane process, Water Res. 226 (2022) 119298, https://doi.org/10.1016/j.watres.2022.119298.
[55] J. Wang, H. Liang, X. Tang, Z. Gan, G. Li, Chemicals-free approach control interface characteristics of nanofiltration membrane: Feasibility and mechanism insight into CEM electrolysis, Water Res. 206 (2021) 117761, https://doi.org/10.1016/j.watres.2021.117761.
[56] Y. Zhao, X. Tong, Y. Chen, Fit-for-Purpose Design of Nanofiltration Membranes for Simultaneous Nutrient Recovery and Micropollutant Removal, Environ. Sci. Technol. 55 (2021) 3352–3361, https://doi.org/10.1021/acs.est.0c08101.
[57] Q. Lu, Z. Wang, J. Wang, L. Xie, Q. Liu, H. Zeng, Deciphering the specific interaction of humic acid with divalent cations at the nanoscale, Chem. Eng. J. 470 (2023) 144097, https://doi.org/10.1016/j.cej.2023.144097.
[58] T. Briskot, N. Hillebrandt, S. Kluters, G. Wang, J. Studts, T. Hahn, T. Huuk, J. Hubbuch, Modeling the Gibbs–Donnan effect during ultrafiltration and diafiltration processes using the Poisson–Boltzmann theory in combination with a basic Stern model, J. Membr. Sci. 648 (2022) 120333, https://doi.org/10.1016/j.memsci.2022.120333.
[59] P. Polanowski, A. Sikorski, The influence of constraints on gelation in a controlling/living copolymerization process, Int. J. Mol. Sci. 24 (2023) 2701, https://doi.org/10.3390/ijms24032701.
[60] B. Ouadi, A. Khatir, E. Magagnini, M. Mokadem, L. Abualigah, A. Smerat, Optimizing silt density index prediction in water treatment systems using pressure-based gradient boosting hybridized with Salp Swarm Algorithm, J. Water Process Eng. 68 (2024) 106479, https://doi.org/10.1016/j.jwpe.2024.106479.
[61] S. Zhu, Q. Zhao, Y. Chen, M.S. Khan, C.-M. Chang, T.-S. Chung, S.B. Chen, Interface-Engineered Nanohybrid Membranes for Selective Boron Removal from Brackish Water and Seawater Reverse Osmosis Permeate, Environ. Sci. Technol. 60 (2026) 2160–2172, https://doi.org/10.1021/acs.est.5c11434.
[62] D.A. Da'na, M.Y. Ashfaq, W.J. Lau, M.A. Al-Ghouti, Dual-functional coatings for RO membranes: optimizing graphene oxide and polydopamine for fouling and scaling control, Molecules 31 (2026) 1702, https://doi.org/10.3390/molecules31101702.
[63] A.A. Abdelkawi, J.L. Lindsey, N.C. Wright, Prediction of gypsum induction time to inform scaling kinetics using machine learning and Smoluchowski theory, Desalination 616 (2025) 119288, https://doi.org/10.1016/j.desal.2025.119288.
[64] K.-G. Lu, S. Ma, D. Hua, H. Liu, C. Li, J. Song, H. Huang, Y. Qin, Silica mitigated calcium mineral scaling in brackish water reverse osmosis, Water Res. 243 (2023) 120428, https://doi.org/10.1016/j.watres.2023.120428.
[65] Y. Li, X. Li, Y. Hao, Y. Liu, Z. Dong, K. Li, Biological and Physiochemical Methods of Biofilm Adhesion Resistance Control of Medical-Context Surface, Int. J. Biol. Sci. 17 (2021) 1769–1781, https://doi.org/10.7150/ijbs.59025.
[66] E. Balducci, F. Papi, D.E. Capialbi, L. Del Bino, Polysaccharides' structures and functions in biofilm architecture of antimicrobial-resistant (AMR) pathogens, Int. J. Mol. Sci. 24 (2023) 4030, https://doi.org/10.3390/ijms24044030.
[67] S. Li, H. Zhang, S. Zhang, S. Chen, L. Zhang, Stress responses of partial denitrification system under long-term ciprofloxacin exposure in an anaerobic

sequencing batch reactor, J. Environ. Chem. Eng. 11 (2023) 110141, https://doi.org/10.1016/j.jece.2023.110141.
[68] B. Biswas, A. Rana, N. Gupta, I. Gupta, R. Puria, A. Thakur, A Novel Robust Method Mimicking Human Substratum To Dissect the Heterogeneity of Candida auris Biofilm Formation, Microbiol. Spectrum 11 (2023) e0089223, https://doi.org/10.1128/spectrum.00892-23.
[69] M. Guźniczak, A. Kublicka, A. Matczuk, A. Wieliczko, I. Buzalewicz, From Structure to Identity: Quantitative Attenuation-Based Optical Biosensing of Bacterial Biofilms and Colonies via Multiparametric OCT, 2026, https://doi.org/10.2139/ssrn.6519461.
[70] B. Dias, J.P.S. Mendes, J.M. Marques Martins de Almeida, L.C.C. Coelho, Simple optical fiber interferometer for dynamic measurement of refractive index and thickness of polymer films, IEEE Sensors J. 22 (2022) 11732–11739, https://doi.org/10.1109/JSEN.2022.3171046.
[71] L. Zhang, B. Cheng, Fractional Fourier Transform and Transferred CNN Based on Tensor for Hyperspectral Anomaly Detection, IEEE Geosci. Remote Sens. Lett. 19 (2022) 1–5, https://doi.org/10.1109/LGRS.2021.3072249.
[72] S.M. Khder, E.A.H. Mohamed, A. Elbialy, I.A. Yassine, Computer aided grading system of digital microscopic blastocyst images, Int. J. Imaging Syst. Technol. 35 (2025) e70162, https://doi.org/10.1002/ima.70162.
[73] T. Jiang, T. Tian, Y.-F. Guan, H.-Q. Yu, Contrasting behaviors of pre-ozonation on ceramic membrane biofouling: Early stage vs late stage, Water Res. 220 (2022) 118702, https://doi.org/10.1016/j.watres.2022.118702.
[74] J.A. Gutiérrez-Gutiérrez, V. Mieites, A. Pontón, I. García-Montesinos, J.M. López-Higuera, O.M. Conde, Evaluation of cardiovascular pathologies in human aortas with hyperspectral imaging and optical coherence tomography, , in: Clinical Biophotonics III,, SPIE, 2024, p. 69, https://doi.org/10.1117/12.3022425.
[75] J. Liu, Z. Li, Y. Wang, X. Liu, G. Tu, W. Li, Analyzing scaling behavior of calcium sulfate in membrane distillation via optical coherence tomography, Water Res. 191 (2021) 116809, https://doi.org/10.1016/j.watres.2021.116809.
[76] Y. Chemla, I. Levin, Y. Fan, A.A. Johnson, C.W. Coley, C.A. Voigt, Hyperspectral reporters for long-distance and wide-area detection of gene expression in living bacteria, Nat. Biotechnol. 44 (2026) 258–268, https://doi.org/10.1038/s41587-025-02622-y.
[77] A. Gündüz, Z. Orman, Enhancing Hyperspectral Image Classification with Bayesian for CNN-GRU Hyperparameter Optimization, in: F.P. Garcia, A. Jamil, A.A. Hameed, A. Ortis, I.S. Ramirez (Eds.), Recent Trends and Advances in Artificial Intelligence, Springer Nature Switzerland, Cham, 2024, pp. 640–652, https://doi.org/10.1007/978-3-031-70924-1_48.
[78] J. Lee, M. Zhan, Y. Kim, S. Hong, Comparison of different cleaning strategies on fouling mitigation in hollow fiber nanofiltration membranes for river water treatment, J. Clean. Prod. 380 (2022) 134764, https://doi.org/10.1016/j.jclepro.2022.134764.
[79] H. Wang, R. Dai, L. Wang, X. Wang, Z. Wang, Membrane fouling behaviors in a full-scale zero liquid discharge system for cold-rolling wastewater brine treatment: A comprehensive analysis on multiple membrane processes, Water Res. 226 (2022) 119221, https://doi.org/10.1016/j.watres.2022.119221.
[80] M.A. Ahmed, S.A. Mahmoud, S. Amin, A.A. Mohamed, The persistent challenge of biofouling in reverse osmosis desalination: a review of characterization, control, and future directions, Water Air Soil Pollut. 236 (2025) 599, https://doi.org/10.1007/s11270-025-08179-5.
[81] Y.-Q. Xu, Z.-Y. Huo, Y. Bai, Y. Wang, W.-L. Wang, H.-B. Wang, Y.-J. Gao, R.-N. Wang, C.-Y. Li, H. Liu, W.E. Huang, H.-Y. Hu, Y.-H. Wu, Isothiazolinone biocide aggravate disinfection-resistant bacterial fouling on reverse osmosis membranes, Desalination 620 (2026) 119637, https://doi.org/10.1016/j.desal.2025.119637.
[82] F.Y. Eguda, A.O. Amoo, S.B. Adamu, O.D. Ogwumu, S.A. Somma, I.B. Babangida, A mathematical model for water quality assessment: evidence-based from selected boreholes in Federal University Dutse, Nigeria, USci 2 (2023) 161–168, https://doi.org/10.56919/usci.2324.020.
[83] C. Shen, W. Mu, C. Wang, W. Xu, P. Hedström, Uncovering the generic and alloy-specific governing parameters of deformation-induced martensitic transformation in austenitic steel, J. Mater. Sci. 59 (2024) 3087–3100, https://doi.org/10.1007/s10853-023-09325-2.
[84] L. Stüwe, A. Lüken, F. Stockmeier, L. Griesberg, T. Kratzenberg, J. Linkhorst, W. Richtering, M. Wessling, Temporal resistance fluctuations during the initial filtration period of colloidal matter filtration, J. Membr. Sci. 708 (2024) 122988, https://doi.org/10.1016/j.memsci.2024.122988.
[85] W. Qin, X. Luo, Asynchronous Parallel Fuzzy Stochastic Gradient Descent for High-Dimensional Incomplete Data Representation, IEEE Trans. Fuzzy Syst. 32 (2024) 445–459, https://doi.org/10.1109/TFUZZ.2023.3300370.
[86] Y. Zhang, Y. Cai, L. Zeng, P. Liu, L.Z. Ma, J. Liu, A microfluidic approach for quantitative study of spatial heterogeneity in bacterial biofilms, Small Sci. 2 (2022) 2200047, https://doi.org/10.1002/smsc.202200047.
[87] Y. He, X. Huang, T. Li, X. Lv, N. Tang, C. Feng, B. Shi, Ultrafiltration membrane fouling control by two-stage coagulant dosing with moderate pH adjustment, Desalination 537 (2022) 115893, https://doi.org/10.1016/j.desal.2022.115893.
[88] F. Ellouze, Y. Kammoun, N. Kalboussi, A. Rapaport, J. Harmand, S. Nasr, N. B. Amar, Optimal control of backwash scheduling for pumping energy saving: Application to the treatment of urban wastewater, J. Water Process Eng. 56 (2023) 104378, https://doi.org/10.1016/j.jwpe.2023.104378.
[89] C. Jiang, Z. Fei, M. Zhang, Z. Liu, X. Liu, Q.J. Niu, Y. Hou, Poly(piperazine-amide) nanofiltration membrane with innate positive charge for enhanced bivalent cation rejection and mono/bivalent cation selectivity, J. Membr. Sci. 664 (2022) 121060, https://doi.org/10.1016/j.memsci.2022.121060.
[90] A. Das, D.M. Warsinger, Batch counterflow reverse osmosis, Desalination 507 (2021) 115008, https://doi.org/10.1016/j.desal.2021.115008.
[91] A. Pervov, D. Spitsov, Production of drinking water with membranes with simultaneous utilization of concentrate and reject effluent after sludge dewatering, Membranes 13 (2023) 133, https://doi.org/10.3390/membranes13020133.
[93] K. Cui, S. Rao, S. Xu, Y. Huang, X. Cai, Z. Huang, Y. Wang, X. Feng, F. Liu, W. Zhang, Y. Li, S. Wang, Spectral convolutional neural network chip for in-sensor edge computing of incoherent natural light, Nat. Commun. 16 (2025) 81, https://doi.org/10.1038/s41467-024-55558-3.
[94] T. Xiao, Z. Zhu, L. Li, J. Shi, Z. Li, X. Zuo, Membrane fouling and cleaning strategies in microfiltration/ultrafiltration and dynamic membrane, Sep. Purif. Technol. 318 (2023) 123977, https://doi.org/10.1016/j.seppur.2023.123977.
[95] T.A. Malkoske, Y.-H. Cai, S.E. Bone, A.I. Schäfer, Calcium-organic matter fouling in nanofiltration: Synchrotron-based X-ray fluorescence and absorption near-edge structure spectroscopy for speciation, Water Res. 272 (2025) 122938, https://doi.org/10.1016/j.watres.2024.122938.
[96] Y. Yao, P. Zhang, F. Sun, W. Zhang, M. Li, G. Sha, L. Teng, X. Wang, M. Huo, R. M. DuChanois, T. Cao, C. Boo, X. Zhang, M. Elimelech, More resilient polyester membranes for high-performance reverse osmosis desalination, Science 384 (2024) 333–338, https://doi.org/10.1126/science.adk0632.
[97] T. Glover, K.I. Kwartei, The Development of Real-Time Fouling Monitoring and Control Systems for Reverse Osmosis Membrane Cleaning using Deep Reinforcement Learning. http://hdl.handle.net/10919/115741, 2023 (accessed June 12, 2026).
[98] T. Zhong, T. Lin, X. Zhang, F. Jiang, H. Chen, Impact of biological activated carbon filtration and backwashing on the behaviour of PFASs in drinking water treatment plants, J. Hazard. Mater. 446 (2023) 130641, https://doi.org/10.1016/j.jhazmat.2022.130641.
[99] B.P. Meneses, A. Amadon, Physical limits to human brain B0 shimming with spherical harmonics, engineering implications thereof, MAGMA 35 (2022) 923–941, https://doi.org/10.1007/s10334-022-01025-3.
[100] F. Brumand-Poor, T. Kotte, A. Hanifa, C. Reese, M. Hofmeister, K. Schmitz, Advancing pressure-based flow rate soft sensors: signal filtering effects and non-laminar flow rate determination, J. Exp. Theor. Anal. 3 (2025) 8, https://doi.org/10.3390/jeta3010008.
[101] X. Che, H.K. Zhang, Z.B. Li, Y. Wang, Q. Sun, D. Luo, H. Wang, Linearly interpolating missing values in time series helps little for land cover classification using recurrent or attention networks, ISPRS J. Photogramm. Remote Sens. 212 (2024) 73–95, https://doi.org/10.1016/j.isprsjprs.2024.04.021.
[102] S. Zhang, J. Li, Z. Mei, P. Xia, Y. Ni, Y. Qian, F. Shu, Consistent Power System Time Series Imputation via Denoising Diffusion Probabilistic Models, IEEE Trans. Smart Grid (2026) 1, https://doi.org/10.1109/TSG.2026.3687898.
[103] H. Pang, Y. Ben, Y. Cao, S. Qu, C. Hu, Time series-based machine learning for forecasting multivariate water quality in full-scale drinking water treatment with various reagent dosages, Water Res. 268 (2025) 122777, https://doi.org/10.1016/j.watres.2024.122777.
[104] X. Chen, M. Liu, Z. Wang, Y. Wang, Explainable deep learning-based feature selection and intrusion detection method on the Internet of Things, Sensors (Basel, Switzerland) 24 (2024) 5223, https://doi.org/10.3390/s24165223.
[105] A. Faqih, M.J. Vikri, I.A. Sa'ida, Optimizing Gated Recurrent Unit (GRU) for gold price prediction: hyperparameter tuning and model evaluation on historical XAU/USD data, Jurnal Sisfokom (Sistem Informasi Dan Komputer) 14 (2025) 141–147, https://doi.org/10.32736/sisfokom.v14i2.2352.
[106] B. Kou, T. Wen, Identification model of mine water inrush source based on XGBoost and SHAP, Sci. Rep. 15 (2025) 1–17, https://doi.org/10.1038/s41598-024-83710-y.
[107] A. Imamura, N. Arizumi, Gabor Filter Incorporated CNN for Compression, in: 2021 36th International Conference on Image and Vision Computing New Zealand (IVCNZ), 2021, pp. 1–5, https://doi.org/10.1109/IVCNZ54163.2021.9653342.
[108] O. Usman, O. Ozkan, A. Koy, T.S. Adebayo, Energy-related uncertainty shocks and inflation dynamics in the U.S: A multivariate quantile-on-quantile regression approach, Struct. Chang. Econ. Dyn. 71 (2024) 235–247, https://doi.org/10.1016/j.strueco.2024.07.012.
[109] A. Mesbah, R. Wood, W. Gao, T. Noël, A.I. Cooper, D. Tilbury, S.J. Qin, Sensing connections in automation, control and robotics, Nat. Chem. Eng. 2 (2025) 281–284, https://doi.org/10.1038/s44286-025-00226-6.
[110] A.J. Rothman, P. Sheeran, The operating conditions framework: Integrating mechanisms and moderators in health behavior interventions, Health Psychol. 40 (2021) 845–857, https://doi.org/10.1037/hea0001026.
[111] O. Samuelsson, A. Björk, B. Carlsson, Model-based monitoring of diffuser fouling using standard sensors, Water Res. X 13 (2021) 100118, https://doi.org/10.1016/j.wroa.2021.100118.
[112] H.-H. Chen, Insuring every action: an authority frontier framework for runtime actuarial control of autonomous AI agents, (2026). doi:10.48550/arXiv.2605.25632.
[113] K. Pan, X. Shen, S. Li, A Utility-Aware Probabilistic Digital Twin Framework With CC-MPC for Hard-Compliance Real-Time Decision Support in Continuous-Flow Environmental Treatment, IEEE Access 14 (2026) 71678–71689, https://doi.org/10.1109/ACCESS.2026.3691971.
[114] B. Niu, L. Yang, S. Meng, D. Liang, H. Liu, L. Yang, L. Shen, Q. Zhao, Time-dependent analysis of polysaccharide fouling by Hermia models: Reveal the structure of fouling layer, Sep. Purif. Technol. 302 (2022) 122093, https://doi.org/10.1016/j.seppur.2022.122093.

[115] M.N. Mangal, V.A. Yangali-Quintanilla, S.G. Salinas-Rodriguez, J. Dusseldorp, B. Blankert, A.J.B. Kemperman, J.C. Schippers, M.D. Kennedy, W.G.J. van der Meer, Application of a smart dosing pump algorithm in identifying real-time optimum dose of antiscalant in reverse osmosis systems. http://hdl.handle.net/10754/679593, 2022 (accessed June 14, 2026).
[116] E.S. Committee, S.J. More, V. Bampidis, D. Benford, C. Bragard, A. Hernandez-Jerez, S.H. Bennekou, T.I. Halldorsson, K.P. Koutsoumanis, C. Lambré, K. Machera, H. Naegeli, S.S. Nielsen, J.R. Schlatter, D. Schrenk, V. Silano, D. Turck, M. Younes, E. Benfenati, A. Crépet, J.D. Te Biesebeek, E. Testai, B. Dujardin, J.L.C. Dorne, C. Hogstrand, Guidance document on scientific criteria for grouping chemicals into assessment groups for human risk assessment of combined exposure to multiple chemicals, EFSA J. 19 (2021) e07033, https://doi.org/10.2903/j.efsa.2021.7033.
[117] D. Kim, J.P. Yun, T. Lee, D. Jung, H.-I. Won, AI-driven multivariate time series forecasting of transmembrane pressure in ultrafiltration: A stationarity-aware approach, J. Water Process Eng. 77 (2025) 108539, https://doi.org/10.1016/j.jwpe.2025.108539.
[118] H.-J. Qin, M. Zhu, S. Ji, R. Wu, Y. Yamamoto, Y. Qin, Y.-Y. Li, Fouling dynamics varied by filtration cycles in high-solid anaerobic membrane bioreactors for sewage sludge treatment, J. Environ. Manag. 392 (2025) 126659, https://doi.org/10.1016/j.jenvman.2025.126659.
[119] C.S. Nnebuo, O. Nir, Backwashable hollow-fiber membranes enable ultrahigh recovery ratio effluent nanofiltration, Sep. Purif. Technol. 372 (2025) 133354, https://doi.org/10.1016/j.seppur.2025.133354.
[120] M. Khedher, J. Awad, E. Donner, B. Drigo, R. Fabris, M. Harris, K. Braun, C.W. K. Chow, Using the Flocculation Index to optimise coagulant dosing during drinking water treatment, J. Water Process Eng. 51 (2023) 103394, https://doi.org/10.1016/j.jwpe.2022.103394.
[121] M. Xu, D. Liu, AI-enabled membrane bioreactors: a review of control architectures and operating-parameter optimization for nitrogen and phosphorus removal, Water 17 (2025) 2899, https://doi.org/10.3390/w17192899.
[122] C. Chen, Y. Zhang, L. Hou, S. Takizawa, Y. Yang, Insights into dynamic evolution of combined scaling-biofouling in reverse osmosis, J. Membr. Sci. 692 (2024) 122295, https://doi.org/10.1016/j.memsci.2023.122295.
[123] Y. Yin, N. Jeong, R. Minjarez, C.A. Robbins, K.H. Carlson, T. Tong, Contrasting Behaviors between Gypsum and Silica Scaling in the Presence of Antiscalants during Membrane Distillation, Environ. Sci. Technol. 55 (2021) 5335–5346, https://doi.org/10.1021/acs.est.0c07190.
[124] S.F. Anis, B.S. Lalia, M. Khair, R. Hashaikeh, N. Hilal, Electro-ceramic self-cleaning membranes for biofouling control and prevention in water treatment, Chem. Eng. J. 415 (2021) 128395, https://doi.org/10.1016/j.cej.2020.128395.
[126] N. Khosrowabadi, K. Hoberg, C. Imdahl, Evaluating human behaviour in response to AI recommendations for judgemental forecasting, Eur. J. Oper. Res. 303 (2022) 1151–1167, https://doi.org/10.1016/j.ejor.2022.03.017.
[127] A. Golabi, A. Erradi, H. Qiblawey, A. Tantawy, A. Bensaid, K. Shaban, Optimal operation of reverse osmosis desalination process with deep reinforcement learning methods, Appl. Intell. 54 (2024) 6333–6353, https://doi.org/10.1007/s10489-024-05452-8.
[128] J. Fang, N. Chen, H. Jiang, D. Zhang, F. Shen, X. Wang, X. He, T.-S. Chua, NExT-Guard: training-free streaming safeguard without token-level labels, (2026). doi:10.48550/arXiv.2603.02219.
[129] S. Gharibian, H. Hazrati, Towards practical integration of MBR with electrochemical AOP: Improved biodegradability of real pharmaceutical wastewater and fouling mitigation, Water Res. 218 (2022) 118478, https://doi.org/10.1016/j.watres.2022.118478.
[130] H.A. Salman, A. Kalakech, A. Steiti, Random forest algorithm overview, Babylon. J. Mach. Learn. 2024 (2024) 69–79, https://doi.org/10.58496/BJML/2024/007.
[131] A. Schell, H. Oberhauser, Nonlinear independent component analysis for discrete-time and continuous-time signals, Ann. Stat. 51 (2023) 487–518, https://doi.org/10.1214/23-AOS2256.
[132] X. Hu, G. Gao, B. Li, W. Wang, F.M. Ghannouchi, A Novel Lightweight Grouped Gated Recurrent Unit for Automatic Modulation Classification, IEEE Wirel. Commun. Lett. 13 (2024) 2135–2139, https://doi.org/10.1109/LWC.2024.3402975.
[133] A. Agarwal, P. Dey, S. Kumar, Sentiment Analysis using Modified GRU, in: Proceedings of the 2022 Fourteenth International Conference on Contemporary Computing, Association for Computing Machinery, New York, NY, USA, 2022, pp. 356–361, https://doi.org/10.1145/3549206.3549270.
[134] K. Liao, X. Shi, L. Sui, W. Holderbaum, W. Xiong, A Novel Long Short-Term Complementary Memory Network Based on Temporal Distribution Matching for Soft Sensing in Industrial Processes, IEEE Trans. Instrum. Meas. 75 (2026) 1–13, https://doi.org/10.1109/TIM.2025.3648002.
[135] Y. Lu, T. Li, J. Su, CNN-GRU-Attention Hybrid Network for Photovoltaic Array Fault Diagnosis, in: Q. Yang (Ed.), The Proceedings of the 12th Frontier Academic Forum of Electrical Engineering (FAFEE2025), Springer Nature, Singapore, 2026, pp. 579–590, https://doi.org/10.1007/978-981-95-4294-9_61.
[136] M. Waqas, U.W. Humphries, A critical review of RNN and LSTM variants in hydrological time series predictions, MethodsX 13 (2024) 102946, https://doi.org/10.1016/j.mex.2024.102946.
[137] R. Cahuantzi, X. Chen, S. Güttel, K. Arai, A Comparison of LSTM and GRU Networks for Learning Symbolic Sequences, in: Intelligent Computing, Springer Nature Switzerland, Cham, 2023, pp. 771–785, https://doi.org/10.1007/978-3-031-37963-5_53.
[138] A. Yunita, M.I. Pratama, M.Z. Almuzakki, H. Ramadhan, E.A.P. Akhir, A. B. Firdausiah Mansur, A.H. Basori, Performance analysis of neural network architectures for time series forecasting: A comparative study of RNN, LSTM, GRU, and hybrid models, MethodsX 15 (2025) 103462, https://doi.org/10.1016/j.mex.2025.103462.
[139] G. Huang, X. Gao, P. Liu, L. Zhou, LSTM-GRU and Multi-Head Attention Based Multivariate Time Series Prediction Model for Electro-Hydraulic Servo Material Fatigue Testing Machine, CMC 87 (2026), https://doi.org/10.32604/cmc.2026.074941.
[140] Z. Zhao, S. Dong, P. Zhao, Z. Liu, L. Ji, W. Li, Predicting transient thermo-hydraulic parameters of fast reactor cores using two-layer decomposition hybrid neural network, Case Stud. Therm. Eng. 81 (2026) 107983, https://doi.org/10.1016/j.csite.2026.107983.
[141] X. Zheng, Y. Zhao, B. Peng, M. Ge, Y. Kong, S. Zheng, Information Filtering Unit-Based Long Short-Term Memory Network for Industrial Soft Sensor Modeling, IEEE Sensors J. 24 (2024) 13530–13544, https://doi.org/10.1109/JSEN.2024.3367909.
[142] P. Zhu, Y. Li, Y. Hu, S. Xiang, Q. Liu, D. Cheng, Y. Liang, MCI-GRU: Stock Prediction Model Based on Multi-Head Cross-Attention and Improved GRU, Neurocomputing 638 (2025) 130168, https://doi.org/10.1016/j.neucom.2025.130168.
[143] A. Gharehbaghi, R. Ghasemlounia, F. Ahmadi, M. Albaji, Groundwater level prediction with meteorologically sensitive Gated Recurrent Unit (GRU) neural networks, J. Hydrol. 612 (2022) 128262, https://doi.org/10.1016/j.jhydrol.2022.128262.
[144] P. Yao, L. Yu, K. Yue, S. Zhan, H. Chen, A Learnable Threshold and Resistance Spiking Neuron for Efficient Spiking Neural Networks Training, in: D.-S. Huang, Q. Zhang, C. Zhang, W. Chen (Eds.), Advanced Intelligent Computing Technology and Applications, Springer Nature, Singapore, 2025, pp. 263–273, https://doi.org/10.1007/978-981-96-9946-9_23.
[145] K.K. Katibi, A.A. Mutalovich, N.M. Nawi, G.M. Iztleuov, M.I. Sataev, M.A. M. Nasir, An insight into real-time monitoring and predictive modelling of ultrafiltration pretreatment for flow rate and fouling mitigation in seawater desalination, Desalin. Water Treat. 323 (2025) 101386, https://doi.org/10.1016/j.dwt.2025.101386.
[147] B. Wang, W. Lu, Y. Zhang, A graph-enabled parametric modeling approach for façade layout generative design, J. Build. Eng. 105 (2025) 112481, https://doi.org/10.1016/j.jobe.2025.112481.
[148] W. Ahmed Khan, S.-H. Chung, S. Qiang Liu, M. Masoud, X. Wen, Smoothing and Matrix Decomposition-Based Stacked Bidirectional GRU Model for Machine Downtime Forecasting, IEEE Trans. Syst. Man. Cybern. Syst. 55 (2025) 7215–7227, https://doi.org/10.1109/TSMC.2025.3582768.
[149] Y. Meng, N. Jiao, Revisiting wastewater pH standards: A policy lever for mitigating coastal acidification and enhancing blue carbon, Innovation 6 (2025), https://doi.org/10.1016/j.xinn.2025.101003.
[150] B. He, S. Qian, T. Li, Modeling product carbon footprint for manufacturing process, J. Clean. Prod. 402 (2023) 136805, https://doi.org/10.1016/j.jclepro.2023.136805.
[151] S. Kosari, M. Parsinejad, A. Mokhtaran, S. Zebardast, Predicted feasibility and economic return of drainage water recycling in an arid region, Agric. Water Manag. 302 (2024) 108983, https://doi.org/10.1016/j.agwat.2024.108983.
[152] Z.H. Foo, D. Rehman, A.T. Bouma, S. Monsalvo, J.H. Lienhard, Lithium Concentration from Salt-Lake Brine by Donnan-Enhanced Nanofiltration, Environ. Sci. Technol. 57 (2023) 6320–6330, https://doi.org/10.1021/acs.est.2c08584.
[153] A. Chakraborty, N. Parashar, D. Kumar Pandey, P. Kumar, U.V. Deokar, J.P. N. Pandey, M.S. Kulkarni, Radiological complexity of nuclear facilities: an information complexity approach to workplace monitoring, J. Radiol. Prot. 44 (2024), https://doi.org/10.1088/1361-6498/ad42a5.
[154] J. Chang, Q. Hu, C. Liu, C.Y. Tang, Optimal covariance matrix estimation for high-dimensional noise in high-frequency data, J. Econ. 239 (2024) 105329, https://doi.org/10.1016/j.jeconom.2022.06.010.
[155] S. Chalichalamala, N. Govindan, R. Kasarapu, An extreme gradient boost based classification and regression tree for network intrusion detection in IoT, Bull. Electr. Eng. Inform. 13 (2024) 1741–1751, https://doi.org/10.11591/eei.v13i3.6843.
[156] C.-T. Yang, Y.-W. Chan, J.-C. Liu, E. Kristiani, C.-H. Lai, Cyberattacks detection and analysis in a network log system using XGBoost with ELK stack, Soft. Comput. 26 (2022) 5143–5157, https://doi.org/10.1007/s00500-022-06954-8.
[157] P. Prathima, An enhanced XGBoost machine learning model to detect fake social media accounts, Jisem 10 (2025) 372–383, https://doi.org/10.52783/jisem.v10i53s.10883.
[158] W. Mchara, G. Cicceri, L. Manai, M. Raissi, H. Albaqami, Hybrid wavelet–transformer–XGBoost model optimized by chaotic billiards for global irradiance forecasting, J. Sens. Actuator Netw. 15 (2026) 12, https://doi.org/10.3390/jsan15010012.
[159] E. Grotti, P.B. Santana, J.G.P. Filho, H.M. Gomes, Multiobjective robust optimization framework based on first and second order Taylor expansion applied to a vehicle suspension design, Optim. Eng. 25 (2024) 637–668, https://doi.org/10.1007/s11081-023-09817-9.
[160] Y. Maanani, C. Pham, Q. Wang, K.K. Nguyen, K. Al-Haddad, XGBoost for multi-fault diagnosis and prediction in permanent magnet synchronous machines, Electronics 15 (2026) 1759, https://doi.org/10.3390/electronics15081759.
[161] Y. Zhuang, Z. Fan, J. Gou, Y. Huang, W. Feng, A importance-based ensemble method using an adaptive threshold searching for feature selection, Expert Syst. Appl. 267 (2025) 126152, https://doi.org/10.1016/j.eswa.2024.126152.
[162] H. Wang, Y. Chen, J. Kang, Z. Ding, H. Zhu, An XGBoost-Based predictive control strategy for HVAC systems in providing day-ahead demand response, Build. Environ. 238 (2023) 110350, https://doi.org/10.1016/j.buildenv.2023.110350.

[163] H. Wang, Z. Zhang, J. Liu, L. Wang, Q. Liu, T. Zhang, O3/H2O2-NF Coupling for treating coal chemical reverse osmosis Concentrate: Organic matter removal and membrane fouling control, Sep. Purif. Technol. 362 (2025) 131682, https://doi.org/10.1016/j.seppur.2025.131682.
[164] S. Zhao, Y. Chen, G. Wu, J. Li, Y. Ren, X. Duan, Investigation on nanofiltration membrane fouling behaviour of cation-induced apam in strontium-bearing mine water, J. Environ. Chem. Eng. 11 (2023) 110940, https://doi.org/10.1016/j.jece.2023.110940.
[165] M. Song, J. Kim, J. Lee, M. Kim, S. Jung, S. Kang, J. Lee, Optimizing coagulant dosing for enhanced sludge settling and subsequent anaerobic digestion, Bioresour. Technol. 442 (2026) 133670, https://doi.org/10.1016/j.biortech.2025.133670.
[166] S. Zhang, W. Gu, Z. Geng, J. Bai, B. Dong, J. Zhao, X. Zhuang, K. Shih, Immobilization of heavy metals in biochar by co-pyrolysis of sludge and CaSiO3, J. Environ. Manag. 326 (2023) 116635, https://doi.org/10.1016/j.jenvman.2022.116635.
[167] N. Das, N. Ojha, S.K. Mandal, Wastewater treatment using plant-derived bioflocculants: green chemistry approach for safe environment, Water Sci. Technol. 83 (2021) 1797–1812, https://doi.org/10.2166/wst.2021.100.
[168] T. Wagner, P. Saha, H. Bruning, H. Rijnaarts, Lowering the fresh water footprint of cooling towers: A treatment-train for the reuse of discharged water consisting of constructed wetlands, nanofiltration, electrochemical oxidation and reverse osmosis, J. Clean. Prod. 364 (2022) 132667, https://doi.org/10.1016/j.jclepro.2022.132667.
[169] R.R. Patel, P.P. Arun, S.K. Singh, M. Singh, Mycobacterial biofilms: Understanding the genetic factors playing significant role in pathogenesis, resistance and diagnosis, Life Sci. 351 (2024) 122778, https://doi.org/10.1016/j.lfs.2024.122778.
[170] M. Pan, H. Li, X. Han, W. Ma, X. Li, Q. Guo, B. Yang, C. Ding, Y. Ma, Effects of hydrodynamic conditions on the composition, spatiotemporal distribution of different extracellular polymeric substances and the architecture of biofilms, Chemosphere 307 (2022) 135965, https://doi.org/10.1016/j.chemosphere.2022.135965.
[171] J. Xu, J. Jiang, Z. Wang, Enhanced spatiotemporal mapping of urban wetland microplastics: An interpretable CNN-GRU approach using satellite imagery and limited samples, Ecotoxicol. Environ. Saf. 303 (2025) 119003, https://doi.org/10.1016/j.ecoenv.2025.119003.
[172] Y. Rao, W. Zhao, Y. Tang, J. Zhou, S.N. Lim, J. Lu, HorNet: Efficient High-Order Spatial Interactions with Recursive Gated Convolutions, Adv. Neural Inf. Proces. Syst. (2022) 10353–10366, 35.
[173] M. Esmaeili, D. Abbasi-Moghadam, A. Sharifi, A. Tariq, Q. Li, Hyperspectral Image Band Selection Based on CNN Embedded GA (CNNeGA), IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 16 (2023) 1927–1950, https://doi.org/10.1109/JSTARS.2023.3242310.
[174] H. Leonard, X. Jiang, S. Arshavsky-Graham, L. Holtzman, Y. Haimov, D. Weizman, S. Halachmi, E. Segal, Shining light in blind alleys: deciphering bacterial attachment in silicon microstructures, Nanoscale Horiz. 7 (2022) 729–742, https://doi.org/10.1039/D2NH00130F.
[175] A. Aasi, S. Ebrahimi Bajgani, O. Mahdavi, Condition-Aware AI for Predictive Maintenance: Dual-Attention CNN-GRU with Per-Regime Scaling, Expert Syst. Appl. 315 (2026) 131582, https://doi.org/10.1016/j.eswa.2026.131582.
[176] Z. Guo, F. Ren, D. Mei, F. Liu, Z. Li, X. Lu, L. Huang, Topology-Adaptive Graph Attention Networks coupled with radial basis functions: A novel framework for hydraulic conductivity inversion in unstructured meshes, Adv. Water Resour. 209 (2026) 105220, https://doi.org/10.1016/j.advwatres.2026.105220.
[177] G.N. Pathak, B. Sanabria, V.D. Caetano, B. Rafiq, B.K. Rao, Evaluation and treatment monitoring of actinic keratosis using line-field confocal optical coherence tomography, Skin Res. Technol. 30 (2024) e13726, https://doi.org/10.1111/srt.13726.
[178] D.G. Nguyen, S. Park, J. Park, D. Kim, J.S. Eo, K. Han, An MPC Approximation Approach for Adaptive Cruise Control With Reduced Computational Complexity and Low Memory Footprint, IEEE Trans. Intell. Veh. 9 (2024) 3154–3167, https://doi.org/10.1109/TIV.2023.3347203.
[179] D. Campos, V. Galvão, M.L. de Rezende, A. Braga, M. Bodini, U.R.V. Aires, R. Yonaba, L. Goliatt, Automated machine learning achieves accurate water quality prediction with reduced parameter requirements, Sci. Rep. 16 (2026) 4431, https://doi.org/10.1038/s41598-025-34448-8.
[180] E. Rustik, E. Viktoriia, A. Tatyana, Harnessing tensor decomposition for high-dimensional machine learning, (2025). doi:10.36227/techrxiv.174417403.38431928/v1.
[181] V. Francés-Belda, A. Solera-Rico, J. Nieto-Centenero, E. Andrés, C. Sanmiguel Vila, R. Castellanos, Toward aerodynamic surrogate modeling based on β-variational autoencoders, Phys. Fluids 36 (2024) 117139, https://doi.org/10.1063/5.0232644.
[182] Y. Wang, D. Yang, F. Bremond, A. Dantcheva, LIA: Latent Image Animator, IEEE Trans. Pattern Anal. Mach. Intell. 46 (2024) 10829–10844, https://doi.org/10.1109/TPAMI.2024.3449075.
[183] W.-T. Chang, C.-H. Kuo, L.-C. Fang, Variational Channel Distribution Pruning and Mixed-Precision Quantization for Neural Network Model Compression, in: 2022 International Symposium on VLSI Design, Automation and Test (VLSI-DAT), 2022, pp. 1–3, https://doi.org/10.1109/VLSI-DAT54769.2022.9768055.
[184] M. Salman, N. Abbas, S.I. ur Rahman, M. Al Alshaikh, A.K.J. Saudagar, An embedded deep learning framework for real-time violence detection and alert generation, Sci. Rep. 16 (2026) 10805, https://doi.org/10.1038/s41598-026-44939-x.
[185] R. Dave, N. Seliya, N. Siddiqui, S. Mao, The benefits of edge computing in healthcare, smart cities, and IoT, J. Comput. Sci. Appl. 9 (2021) 23–34, https://doi.org/10.12691/jcsa-9-1-3.
[186] L. Zhou, X. Li, Interpretable power load forecasting model based on CNN GRU and SHAP, in: 2023 8th International Conference on Intelligent Informatics and Biomedical Sciences (ICIIBMS), 2023, pp. 569–574, https://doi.org/10.1109/ICIIBMS60103.2023.10347636.
[187] G. Di Marco, D. Lombardi, T. Dounas, A. Agkathidis, A Theoretical Framework for the Standardisation and Industrialisation of 3D Concrete Printing through Decentralised Science: 43rd Conference on Education and Research in Computer Aided Architectural Design in Europe, eCAADe 2025, Proceedings of the 43rd Conference on Education and Research in Computer Aided Architectural Design in Europe, eCAADe 2025, 2025, pp. 555–564, https://doi.org/10.52842/conf.ecaade.2025.1.555.
[188] X. Qin, Y. Chen, B. Li, S. Zeng, G. Mei, C. Yu, A multi-task learning model based on DTP-MMoE for identification of olive oil multi-adulteration using Raman spectroscopy, Foods 15 (2026) 2030, https://doi.org/10.3390/foods15112030.
[189] H.W. Schroer, C.L. Just, Feature Engineering and Supervised Machine Learning to Forecast Biogas Production during Municipal Anaerobic Co-Digestion, ACS EST Eng. 4 (2024) 660–672, https://doi.org/10.1021/acsestengg.3c00435.
[190] Y. Zhang, B. Han, Q. Song, S. Jin, Z. Lu, S. Lu, X. Zhang, Y. Zhang, D. Li, Revealing key structural and operating parameters on salt/dye separation of loose nanofiltration membrane by ensemble machine learning, J. Membr. Sci. 732 (2025) 124274, https://doi.org/10.1016/j.memsci.2025.124274.
[191] A.-L. Balogun, A. Tella, Modelling and investigating the impacts of climatic variables on ozone concentration in Malaysia using correlation analysis with random forest, decision tree regression, linear regression, and support vector regression, Chemosphere 299 (2022) 134250, https://doi.org/10.1016/j.chemosphere.2022.134250.
[192] A. Balla, M.H. Habaebi, E.A.A. Elsheikh, M.R. Islam, F.M. Suliman, The effect of dataset imbalance on the performance of SCADA intrusion detection systems, Sensors (Basel) 23 (2023) 758, https://doi.org/10.3390/s23020758.
[193] S. Mahjoub, L. Chrifi-Alaoui, B. Marhic, L. Delahoche, Predicting energy consumption using LSTM, multi-layer GRU and drop-GRU neural networks, Sensors (Basel, Switzerland) 22 (2022) 4062, https://doi.org/10.3390/s22114062.
[194] M.-W. Li, D.-Y. Xu, J. Geng, W.-C. Hong, A hybrid approach for forecasting ship motion using CNN–GRU–AM and GCWOA, Appl. Soft Comput. 114 (2022) 108084, https://doi.org/10.1016/j.asoc.2021.108084.
[195] J. Cao, A. Xue, Y. Yang, R. Lu, X. Hu, L. Zhang, W. Cao, G. Cao, X. Geng, L. Wang, A hybrid deep learning framework for predicting industrial wastewater influent quality based on graph optimisation, J. Water Process Eng. 65 (2024) 105831, https://doi.org/10.1016/j.jwpe.2024.105831.
[196] J. Engelberg, U. Levy, Generalized metric for broadband flat lens performance comparison, Nanophotonics 11 (2022) 3559–3574, https://doi.org/10.1515/nanoph-2022-0196.
[197] A. Alkatheeri, R. Rafay, E. Alhseinat, A. Safieh, F. Alnaimat, Robust Spatial Modeling of Thermodynamic Parameters in a Full-Scale Reverse Osmosis Membrane Channel, ACS Omega 6 (2021) 12392–12409, https://doi.org/10.1021/acsomega.0c04412.
[198] R.W. Field, J.J. Wu, A.G. Fane, Revisiting the distinct concepts of critical flux and threshold flux, J. Membr. Sci. 741 (2026) 124961, https://doi.org/10.1016/j.memsci.2025.124961.
[199] Z. Bao, X. Li, M. Shi, Y. Huang, J. Du, Strictly chronological CNN embeddings with gradient-boosted trees for next-day log-return forecasting, Symmetry 18 (2026) 416, https://doi.org/10.3390/sym18030416.
[200] H. Chen, X. Liu, D. Gong, C. Zhu, G. Liu, J. Fan, P. Wu, Z. Li, Y. Pan, G. Shi, Y. Sun, G. Zeng, Ultrahigh-water-flux desalination on graphdiyne membranes, Nat. Water 1 (2023) 800–807, https://doi.org/10.1038/s44221-023-00123-3.
[201] J.M. Gorriz, R. Martin-Clemente, F. Segovia, J. Ramirez, A. Ortiz, J. Suckling, Is K-fold cross validation the best model selection method for machine learning? Inf. Fusion 135 (2026) 104404 https://doi.org/10.1016/j.inffus.2026.104404.
[202] W. Zheng, Y. Chen, X. Xu, X. Peng, Y. Niu, P. Xu, T. Li, Research on the factors influencing nanofiltration membrane fouling and the prediction of membrane fouling, J. Water Process Eng. 59 (2024) 104876, https://doi.org/10.1016/j.jwpe.2024.104876.
[203] F.K. Borszcz, A. Ferreira Tramontin, R.D. de Lucas, V. Pereira Costa, Is the 5-Minute Time-Trial Cycling Test a Valid Predictor of Maximal Oxygen Uptake? An External Cross-Validation Study, Int. J. Sports Physiol. Perform. 19 (2024) 565–575, https://doi.org/10.1123/ijspp.2023-0330.
[204] I. Covert, S. Lundberg, S.-I. Lee, Explaining by removing: a unified framework for model explanation, J. Mach. Learn. Res. 22 (2021) 1–90.
[205] T.A. Rainy, M.A. Siddiquee, A systematic review of AI-enabled predictive quality control in advanced metal manufacturing systems, Am. J. Interdiscip. Stud. 7 (2026) 426–458, https://doi.org/10.63125/n73j2x72.
[206] P. Zhang, Z. Xiao, Y. Chen, S. Nie, W. Liang, W. Zhong, T. Yuan, H. Li, W. Zhang, Machine learning-based response and prediction analysis for gravity-driven membrane, Desalination 618 (2026) 119450, https://doi.org/10.1016/j.desal.2025.119450.
[207] B. Liao, L. Kong, W. Lu, J. Zhang, C. Shen, Y. Zhao, J. García, Y. Li, B. Li, S. Cheng, Revealing the role of scale in constructed wetlands: Machine learning approach to design optimization, J. Clean. Prod. 560 (2026) 148324, https://doi.org/10.1016/j.jclepro.2026.148324.
[208] S. Alghaffari, M. Irfan, Z. Mushtaq, N.A. Khan, H. Aziz, A. Shaher, F. Althobiani, Design of a novel adaptive residual fusion technique for SCADA-based wind

turbine anomaly detection, Nondestruct. Test. Eval. 0 (2025) 1–25, https://doi.org/10.1080/10589759.2025.2572405.
[209] J.C.G. Delgado, N. Kuzhagaliyeva, I. Gorbatenko, M. Sarathy, Predicting Vehicle Engine Performance: Assessment of Machine Learning Techniques and Data Imputation, SAE International, 2024, https://doi.org/10.4271/2024-01-2016.
[210] I. Tsaklis, M.J. Gräfe, V. Pettas, Data-driven models for predicting structural loads in offshore wind turbines using SCADA data, J. Phys. Conf. Ser. 3224 (2026) 062042, https://doi.org/10.1088/1742-6596/3224/6/062042.
[211] X. Zhou, Y. Man, S. Liu, J. Zhang, R. Yuan, W. Wang, K. Su, Leveraging multi-level correlations for imputing monitoring data in water supply systems using graph signal sampling theory, Water Res. X 25 (2024) 100274, https://doi.org/10.1016/j.wroa.2024.100274.
[212] W. van Loon, M. Fokkema, F. de Vos, M. Koini, R. Schmidt, M. de Rooij, Imputation of missing values in multi-view data, Inf. Fusion 111 (2024) 102524, https://doi.org/10.1016/j.inffus.2024.102524.
[213] M.W. Heymans, J.W.R. Twisk, Handling missing data in clinical research, J. Clin. Epidemiol. 151 (2022) 185–188, https://doi.org/10.1016/j.jclinepi.2022.08.016.
[214] N. Martindale, T.G. Ellis, M. Jacob, L. Hammes, G. Eshelman, Investigation of impact of split treatment on finished water quality for the Ames water treatment plant, Water Environ. Res. 97 (2025) e70019, https://doi.org/10.1002/wer.70019.
[215] S.P. Sari, A.P. Pratama, R. Herliansyah, F. Novri, A Bayesian MEWMA control chart for monitoring water quality in public utilities, J. Ind. Serv. 11 (2025) 321–340, https://doi.org/10.62870/jiss.v11i2.33473.
[216] B. Karamnia, F. Pourasgharian Roudsari, M.R. Mehrnia, Unsteady state municipal wastewater treatment: MBR response to organic and hydraulic shock-loads, Urban Water J. 21 (2024) 498–507, https://doi.org/10.1080/1573062X.2024.2312513.
[217] D.M.V.P. Ferreira, P.M.S. Carvalho, M.D. Ilić, A Kalman filter approach for state estimation in weakly monitored active distribution networks, Sustain. Energy Grids Netw. 39 (2024) 101423, https://doi.org/10.1016/j.segan.2024.101423.
[218] A. Rivero-Falcón, Y. López-López, B. Peñate Suárez, N. Melián-Martel, Performance evaluation of nanofiltration membranes for SWRO brine valorisation: Insights from a pilot plant under real conditions, Sep. Purif. Technol. 359 (2025) 130774, https://doi.org/10.1016/j.seppur.2024.130774.
[219] D. Xu, X. Zhang, X. Peng, G. Yang, J. Chen, An anti-vibration-shock inertial matching measurement method for hull deformation, IET Signal Process. 16 (2022) 490–500, https://doi.org/10.1049/sil2.12122.
[220] R.C. Pereira, P.H. Abreu, P.P. Rodrigues, Siamese Autoencoder Architecture for the Imputation of Data Missing Not at Random, J. Comput. Sci. 78 (2024) 102269, https://doi.org/10.1016/j.jocs.2024.102269.
[221] S. Daenens, T. Verstraeten, P.-J. Daems, A. Nowé, J. Helsen, Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data, Wind Energ. Sci. 10 (2025) 1137–1152, https://doi.org/10.5194/wes-10-1137-2025.
[222] B. Dickson, J. Mochizuki-Freeman, M.R. Kabir, Z. Tiganj, Time-Local Transformer, Comput. Brain Behav. 9 (2026) 78–90, https://doi.org/10.1007/s42113-025-00253-9.
[223] A.D. Mangussi, R.C. Pereira, A.C. Lorena, P.H. Abreu, Large language models for missing data imputation: understanding behavior, hallucination effects, and control mechanisms, (2026). doi:10.48550/arXiv.2603.22332.
[224] K.F. Grosser, S.C. Lotspeich, T.P. Garcia, Mission imputable: correcting for Berkson error when imputing a censored covariate, (2023). doi:10.48550/arXiv.2303.01602.
[225] S.V.V.S.N. Pichika, P.S.M. Surabhi, S.S.P. Yarramsetty, S. Geetha Rajasekharan, A. Malapati, Missing data imputation and fault classification using generative and transfer learning techniques for a wind turbine gearbox with sensor malfunctions, Struct. Health Monit. (2025) 14759217251370325, https://doi.org/10.1177/14759217251370325.
[226] Z. Wang, R. Sadiq, D. Gan, A Reduced-Order Robust Wide-Area Damping Control for Wind-PV-Thermal-Bundled Power System Considering Operational Uncertainties and Communication Resilience, IEEE Access 12 (2024) 36841–36852, https://doi.org/10.1109/ACCESS.2024.3370677.
[227] S. Sun, J. Wang, Y. Xiao, J. Peng, X. Zhou, Few-shot RUL prediction for engines based on CNN-GRU model, Sci. Rep. 14 (2024) 16041, https://doi.org/10.1038/s41598-024-66377-3.
[228] M. Abouhawwash, Machine Learning-Based Enhanced Embryo Selection Using Temporal Feature Extraction and Rule-Based Labeling Model With Decision Trees, IEEE Access 13 (2025) 212213–212230, https://doi.org/10.1109/ACCESS.2025.3644768.
[229] A.A. Farhan, A. Basharat, N. Allheeib, S. Kanwal, Enhancing smartphone security with human centric bimodal fallback authentication leveraging sensors, Sci. Rep. 14 (2024) 1–11, https://doi.org/10.1038/s41598-024-74473-7.
[230] R. Aruna, R. Madavan, A. Karthick, V. Mohanavel, Black-box modeling and smart control strategies for proton exchange membrane fuel cell stability, Electr. Power Syst. Res. 254 (2026) 112574, https://doi.org/10.1016/j.epsr.2025.112574.
[231] Y. Zhu, B. Lian, Y. Wang, C. Miller, C. Bales, J. Fletcher, L. Yao, T.D. Waite, Machine learning modelling of a membrane capacitive deionization (MCDI) system for prediction of long-term system performance and optimization of process control parameters in remote brackish water desalination, Water Res. 227 (2022) 119349, https://doi.org/10.1016/j.watres.2022.119349.
[233] W. Tingting, W. Lijun, W. Zhongpeng, The practice of reinforcement learning algorithm in dynamic regulation and energy efficiency optimization of wastewater treatment system in industrial parks, Aust. J. Electr. Electron. Eng. 0 (2025) 1–23, https://doi.org/10.1080/1448837X.2025.2553950.
[234] B. Shickel, T.J. Loftus, M. Ruppert, G.R. Upchurch, T. Ozrazgat-Baslanti, P. Rashidi, A. Bihorac, Dynamic predictions of postoperative complications from explainable, uncertainty-aware, and multi-task deep neural networks, Sci. Rep. 13 (2023) 1224, https://doi.org/10.1038/s41598-023-27418-5.
[235] Y. Zhang, C. You, Y. Pei, F. Yang, D. Li, Y. Jiang, Z. Shao, Additional file 1 of Integration of radiogenomic features for early prediction of pathological complete response in patients with triple-negative breast cancer and identification of potential therapeutic targets, 2022, https://doi.org/10.6084/m9.figshare.20025225.v1.
[236] J. Lei, R. Qi, Y. Yang, C. Feng, Targeted stepwise pre-adsorption for enhanced nanofiltration removal of dissolved organic pollutants in petrochemical wastewater, Water Res. 287 (2025) 124390, https://doi.org/10.1016/j.watres.2025.124390.
[237] R. Zafar, Z. Arshad, Y.K. Lee, M.-S. Kim, J. Hur, Unveiling the hidden roles of microplastic-derived dissolved organic matter in membrane fouling using fluorescence and stable carbon isotope tracers, Water Res. 291 (2026) 125165, https://doi.org/10.1016/j.watres.2025.125165.
[238] Z. Han, J. Li, Z. Chen, W. Liu, An improved CFD particulate fouling simulation method: The local fouling method, Int. Commun. Heat Mass Transf. 154 (2024) 107439, https://doi.org/10.1016/j.icheatmasstransfer.2024.107439.
[239] H. Wang, Q. Liang, J.T. Hancock, T.M. Khoshgoftaar, Feature selection strategies: a comparative analysis of SHAP-value and importance-based methods, J. Big Data 11 (2024) 1–16, https://doi.org/10.1186/s40537-024-00905-w.
[240] L. Zhang, L. Jiang, Game-theoretic SHAP-driven interpretable forecasting of air cargo demand using Bayesian-optimized random forests, Front. Phys. 13 (2025), https://doi.org/10.3389/fphy.2025.1705687.
[241] X. Chen, C. Zhao, J. Chen, H. Jiang, D. Li, J. Zhang, B. Han, S. Chen, C. Wang, Water quality parameters-based prediction of dissolved oxygen in estuaries using advanced explainable ensemble machine learning, J. Environ. Manag. 380 (2025) 125146, https://doi.org/10.1016/j.jenvman.2025.125146.
[242] J. Contreras, A. Winterfeld, J. Popp, T. Bocklitz, Spectral Zones-Based SHAP/LIME: Enhancing Interpretability in Spectral Deep Learning Models Through Grouped Feature Analysis, Anal. Chem. 96 (2024) 15588–15597, https://doi.org/10.1021/acs.analchem.4c02329.
[243] Y. Lu, Y. Bu, B. Li, Y. Dai, S. Wang, L. Shi, B. Ma, From black-box prediction to probabilistic control: An explainable CNN-SHAP-Monte Carlo framework for low-carbon wastewater treatment, J. Water Process Eng. 79 (2025) 109050, https://doi.org/10.1016/j.jwpe.2025.109050.
[244] O. Nwafor, C. Nwafor, A. Zakaria, N. Nwankwo, A feed-forward artificial intelligence pipeline for sustainable desalination under climate uncertainties: UAE insights, (2025). doi:10.48550/arXiv.2507.10609.
[245] C. Duckworth, M.J. Guy, A. Kumaran, A.A. O'Kane, A. Ayobi, A. Chapman, P. Marshall, M. Boniface, Explainable machine learning for real-time hypoglycemia and hyperglycemia prediction and personalized control recommendations, J. Diabetes Sci. Technol. 18 (2024) 113–123, https://doi.org/10.1177/19322968221103561.
[246] S.M. Moedt, K.S. Christoffersen, A. Westergaard-Nielsen, K.T. Martinsen, A. Pastor, N.J. Korsgaard, T. Riis, Drivers of epilithic biofilms in Greenland streams: the role of nutrients, temperature and catchment slope across a climate gradient, Environ. Microbiol. Rep. 17 (2025) e70074, https://doi.org/10.1111/1758-2229.70074.
[247] D. Öztürk, F. Bernardi, N.G. Cogan, K. Gupta, S. Chellam, Optimization of surface water microfiltration with hydraulic and chemically enhanced backwashing, J. Membr. Sci. 717 (2025) 123550, https://doi.org/10.1016/j.memsci.2024.123550.
[249] X. Lei, Z. Sun, A. Wang, T. Guo, T. Nagayama, Estimation of bridge girder cumulative displacement for component operational warning using Bayesian neural networks, Struct. Control. Health Monit. 2025 (2025) 9974584, https://doi.org/10.1155/stc/9974584.
[250] D. Gümüş, A. Alver, F. Akbal, Development of a digital twin framework for hybrid adsorption-ultrafiltration systems in drinking water treatment, Process. Saf. Environ. Prot. 205 (2026) 108182, https://doi.org/10.1016/j.psep.2025.108182.
[251] W. Liu, Y. He, Y. Shu, L. Chen, F. Zan, X. Lu, J. Mao, X. Wu, An effluent risk informed closed-loop framework for early warning of influent anomalies using COD soft sensing, Water Res. 299 (2026) 125893, https://doi.org/10.1016/j.watres.2026.125893.
[252] Z. Zhang, X. Li, H. Liu, T. Zhou, Z. Wang, L.D. Nghiem, Q. Wang, Biofouling control of reverse osmosis membrane using free ammonia as a cleaning agent, J. Membr. Sci. 694 (2024) 122414, https://doi.org/10.1016/j.memsci.2024.122414.
[253] A.M. Antoniadi, Y. Du, Y. Guendouz, L. Wei, C. Mazo, B.A. Becker, C. Mooney, Current challenges and future opportunities for XAI in machine learning-based clinical decision support systems: a systematic review, Appl. Sci. 11 (2021) 5088, https://doi.org/10.3390/app11115088.
[254] A. Riyadh, Applications of machine learning for real-time water quality monitoring in water distribution systems, University of British Columbia, 2025, https://doi.org/10.14288/1.0448699.
[255] S. Hong, W. Yu, T. Chai, A Human-in-the-Loop Framework for Interactive and Explainable Data-Driven Modeling, IEEE Trans. Emerg. Top. Comput. Intell. 10 (2026) 1072–1083, https://doi.org/10.1109/TETCI.2025.3637807.
[256] M. Rosa-Morales, M. Ravichandran, W. Song, M. Yazdani-Asrami, Physically interpretable and AI-powered applied-field thrust modelling for magnetoplasmadynamic space thrusters using symbolic regression: towards more explainable predictions, Aerospace 13 (2026) 245, https://doi.org/10.3390/aerospace13030245.
[257] H. John, S.K. Giri, L.K. Sinha, R. Pandiselvam, M. Sain, A. Ray, Investigating flux decline and fouling mechanisms in ultrafiltration of soy protein using a 10 kDa

hollow fiber membrane module, J. Food Process Eng. 48 (2025) e70113, https://doi.org/10.1111/jfpe.70113.
[258] P. Wang, G. An, I. Carra, F. Hassard, P. Campo Moreno, H. Sakar, M. Jodkowska, D. Wang, B. Jefferson, W. Chu, P. Jarvis, Removal of perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS) by coagulation: Influence of coagulant and dosing conditions, Sep. Purif. Technol. 355 (2025) 129562, https://doi.org/10.1016/j.seppur.2024.129562.
[259] M.N. Mangal, S.G. Salinas-Rodriguez, J. Dusseldorp, A.J.B. Kemperman, J. C. Schippers, M.D. Kennedy, W.G.J. van der Meer, Effectiveness of antiscalants in preventing calcium phosphate scaling in reverse osmosis applications, J. Membr. Sci. 623 (2021) 119090, https://doi.org/10.1016/j.memsci.2021.119090.
[260] K.T. Huisman, N. Franco-Clavijo, J.S. Vrouwenvelder, B. Blankert, Improved quantitative evaluation of the fouling potential in spacer-filled membrane filtration channels through a biofouling index based on the relative pressure drop. http://hdl.handle.net/10754/687817, 2023 (accessed June 17, 2026).
[261] Y. Sun, Y. Zheng, A method of gas sensor drift compensation based on intrinsic characteristics of response curve, Sci. Rep. 13 (2023) 1–9, https://doi.org/10.1038/s41598-023-39246-8.
[262] Y. Zhang, Z. Gao, Adaptive sample splitting for randomization tests. https://www.semanticscholar.org/paper/Adaptive-sample-splitting-for-randomization-tests-Zhang-Gao/c90495c0b55bce0a7dd9aea0af8ade22ae391587, 2025. (Accessed 20 May 2026).
[263] S.K. Garg, M. Kipnes, K. Castorino, T.S. Bailey, H.K. Akturk, J.B. Welsh, M. P. Christiansen, A.K. Balo, S.A. Brown, J.L. Reid, S.E. Beck, Accuracy and Safety of Dexcom G7 Continuous Glucose Monitoring in Adults with Diabetes, Diabetes Technol. Ther. 24 (2022) 373–380, https://doi.org/10.1089/dia.2022.0011.
[264] T. Johansen, I.B. Utne, Human-autonomy collaboration in supervisory risk control of autonomous ships, J. Mar. Eng. Technol. 23 (2024) 135–153, https://doi.org/10.1080/20464177.2024.2319369.
[265] C. Song, Y. Shi, M. Li, L. Wu, X. Xiong, J. Liu, D. Xia, An efficient catalyst screening strategy combining machine learning and causal inference, J. Environ. Manag. 377 (2025) 124665, https://doi.org/10.1016/j.jenvman.2025.124665.
[266] X. Zhu, C. Fan, Y. Fang, W. Yu, Y. Xie, H. Liu, Fouling and chemical cleaning strategies for submerged ultrafiltration membrane: synchronized bench-scale, full-scale, and engineering tests, Membranes 14 (2024) 251, https://doi.org/10.3390/membranes14120251.
[267] S.V. Buggenhout, The open membrane database: Synthesis–structure–performance relationships of polymeric membranes, in: KU Leuven Open Science Day, 2023, https://doi.org/10.21428/1192f2f8.5d15580e.
[268] C. Schissel, D.T. Allen, Impact of the High-Emission Event Duration and Sampling Frequency on the Uncertainty in Emission Estimates, Environ. Sci. Technol. Lett. 9 (2022) 1063–1067, https://doi.org/10.1021/acs.estlett.2c00731.
[269] C. Wang, Z. Liu, J. Fei, Z. Long, P. Wang, T. Yu, Evaluating the generalizability and transferability of acoustic leak detection models for water distribution networks, Water Res. 286 (2025) 124273, https://doi.org/10.1016/j.watres.2025.124273.
[270] Q. Wang, W. Lin, S. Chou, P. Dai, X. Huang, Patterned membranes for improving hydrodynamic properties and mitigating membrane fouling in water treatment: A review, Water Res. 236 (2023) 119943, https://doi.org/10.1016/j.watres.2023.119943.
[271] F. Gao, J. Zhou, H. Jia, Q. Wang, J. Wang, Dynamic Monitoring of Reverse Osmosis Process via Electrical Impedance Tomography: Enabling In Situ Visualization and Uncovering Distinct Roles of Concentration Polarization/Foulant Deposition in Flux Decline, Environ. Sci. Technol. 60 (2026) 4430–4439, https://doi.org/10.1021/acs.est.5c15320.
[272] R. Navarro-Tovar, G.A. Zoumpouli, P. Gorgojo, P. Martin, Y.M.J. Chew, D. Mattia, M. Perez-Page, Mitigation of organic fouling in membrane distillation via 3D printed wavy composite membranes, Desalination 614 (2025) 119201, https://doi.org/10.1016/j.desal.2025.119201.
[273] W.S. Al-Rashed, Wastewater membrane bioreactors: a comprehensive review of explainable artificial intelligence and digital twin applications, Membranes 16 (2026) 181, https://doi.org/10.3390/membranes16050181.
[274] K. Gao, Y. Yang, A. Li, J. Pu, S. Takizawa, N.J.D. Graham, L. Hou, Fouling behavior of BTEX in petrochemical wastewater treated by nanofiltration (NF), J. Hazard. Mater. 476 (2024) 135185, https://doi.org/10.1016/j.jhazmat.2024.135185.